

JAVA PROGRAMMING LAB (BCS-351)

B.Tech CSE 3rd SEMESTER

SUBMITTED TO:

MR. UNWAN KHAN

SUBMITTED BY:

KARTIKEYA SINGH

Roll No: BCS2025378

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Assignment 1: Hello World, Variables and Arithmetic

Assignment Aim / Objective: Print text, store values in variables and calculate with arithmetic operators.

Question 1

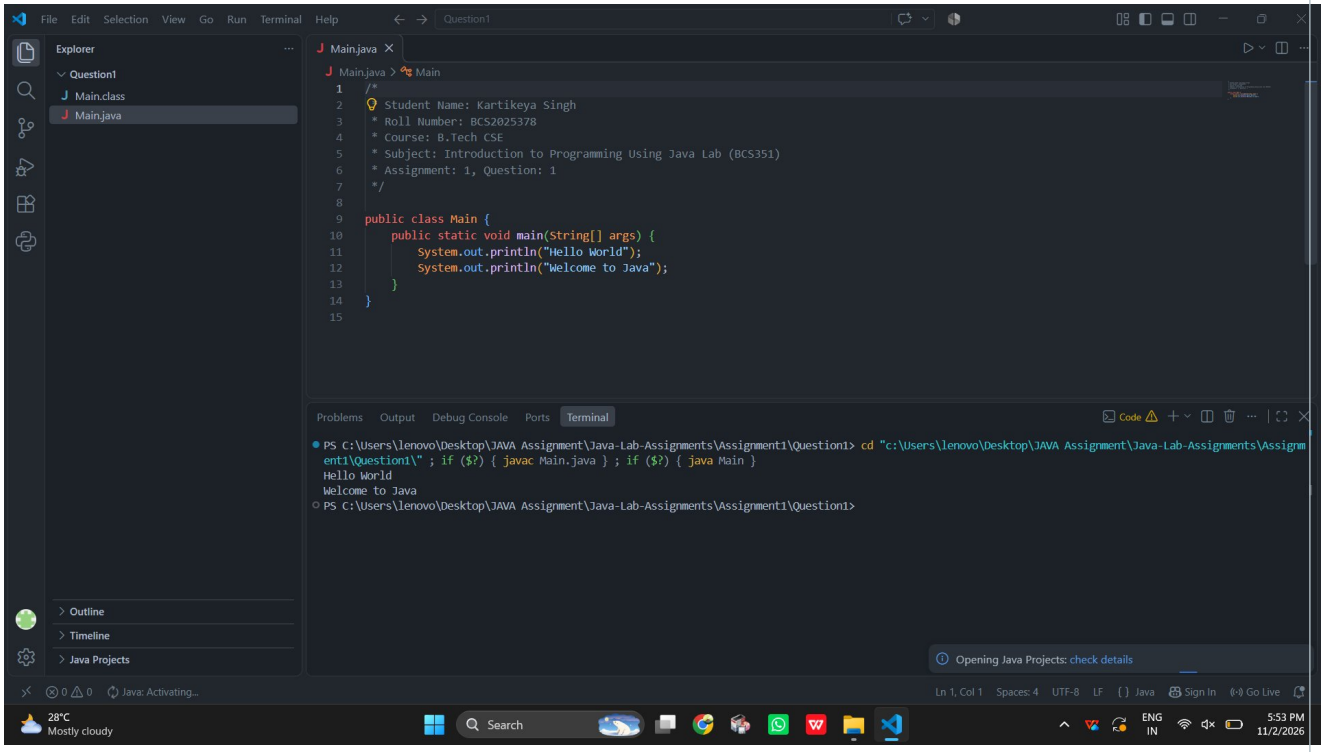
Problem Statement: Print Hello World on one line. Then print Welcome to Java on the next line.

Source Code (Main.java):

```
/*
 * Student Name: Kartikeya Singh
 * Roll Number: BCS2025378
 * Course: B.Tech CSE
 * Subject: Introduction to Programming Using Java Lab (BCS351)
 * Assignment: 1, Question: 1
 */

public class Main {
    public static void main(String[] args) {
        System.out.println("Hello World");
        System.out.println("Welcome to Java");
    }
}
```

Execution Output Screenshot:



Question 2

Problem Statement: Create int roll = 101, double percentage = 84.5, char section = 'B' and String name = "Asha". Print each value on its own line, after the labels Roll:, Percentage:, Section: and Name: in that order.

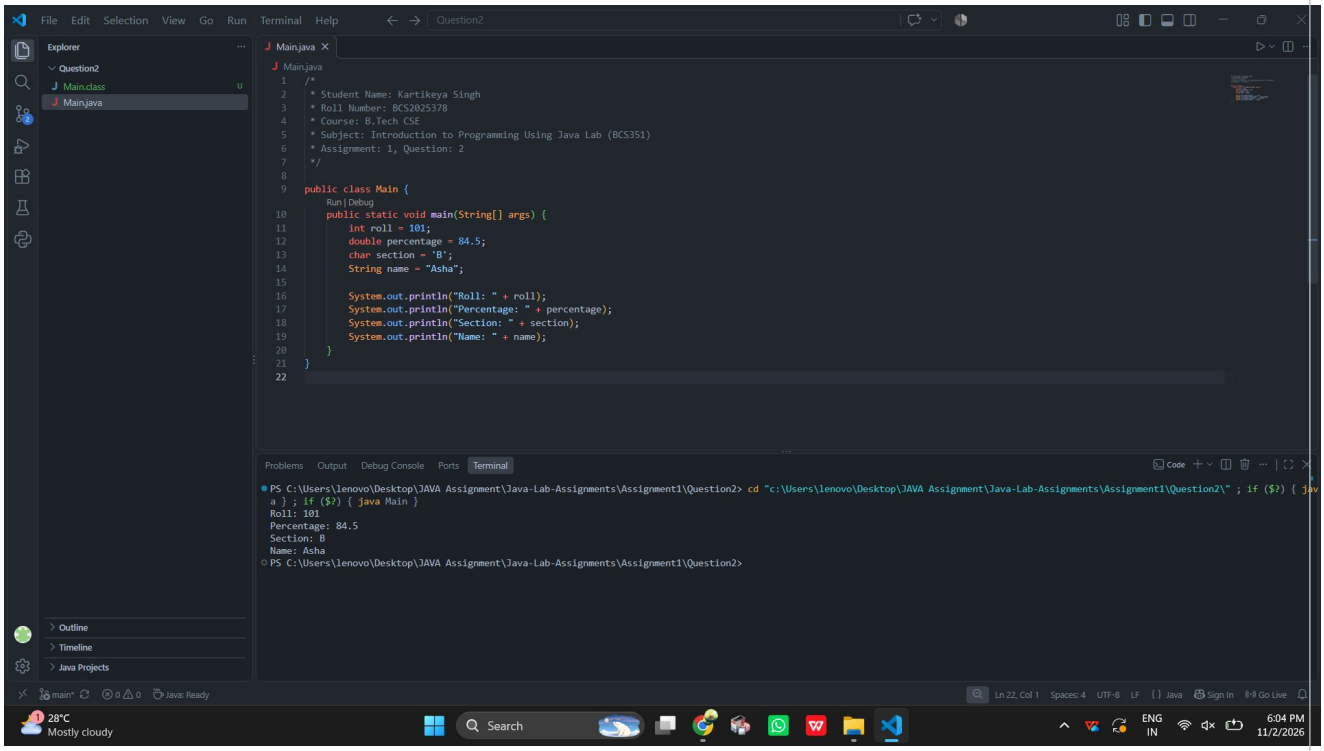
Source Code (Main.java):

```
/*
 * Student Name: Kartikeya Singh
 * Roll Number: BCS2025378
 * Course: B.Tech CSE
 * Subject: Introduction to Programming Using Java Lab (BCS351)
 * Assignment: 1, Question: 2
 */

public class Main {
    public static void main(String[] args) {
        int roll = 101;
        double percentage = 84.5;
        char section = 'B';
        String name = "Asha";

        System.out.println("Roll: " + roll);
        System.out.println("Percentage: " + percentage);
        System.out.println("Section: " + section);
        System.out.println("Name: " + name);
    }
}
```

Execution Output Screenshot:



Question 3

Problem Statement: Create int variables a = 17 and b = 5. Print Sum:, Difference:, Product:, Quotient: and Remainder:, each followed by its result for a + b, a - b, a * b, a / b and a % b.

Source Code (Main.java):

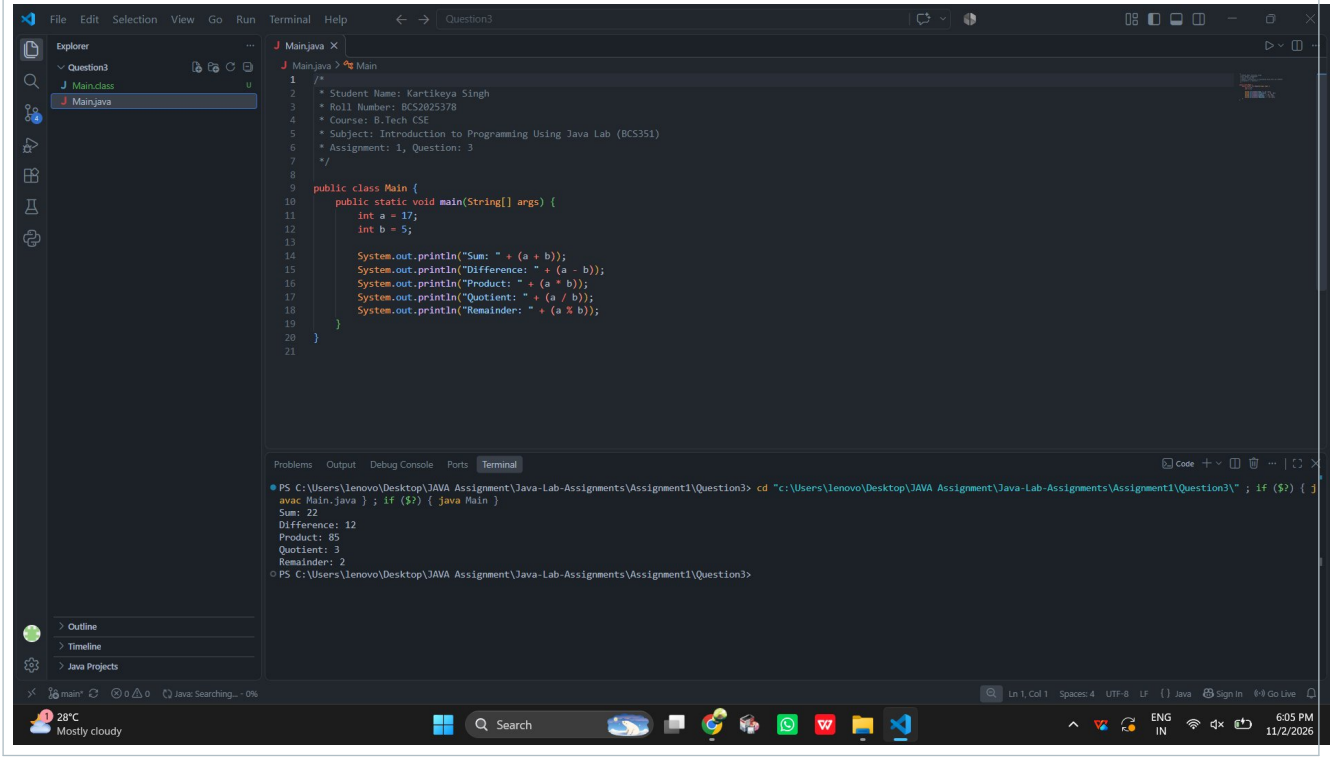
```
/*
 * Student Name: Kartikeya Singh
 * Roll Number: BCS2025378
 * Course: B.Tech CSE
 * Subject: Introduction to Programming Using Java Lab (BCS351)
 * Assignment: 1, Question: 3
 */

public class Main {
    public static void main(String[] args) {
        int a = 17;
        int b = 5;

        System.out.println("Sum: " + (a + b));
        System.out.println("Difference: " + (a - b));
        System.out.println("Product: " + (a * b));
        System.out.println("Quotient: " + (a / b));
        System.out.println("Remainder: " + (a % b));
    }
```

```
}
```

Execution Output Screenshot:



Question 4

Problem Statement: Create a constant named PI with the final keyword and the value 3.14. For a circle with radius 7, print Area: followed by the area and Perimeter: followed by the perimeter, using PI.

Source Code (Main.java):

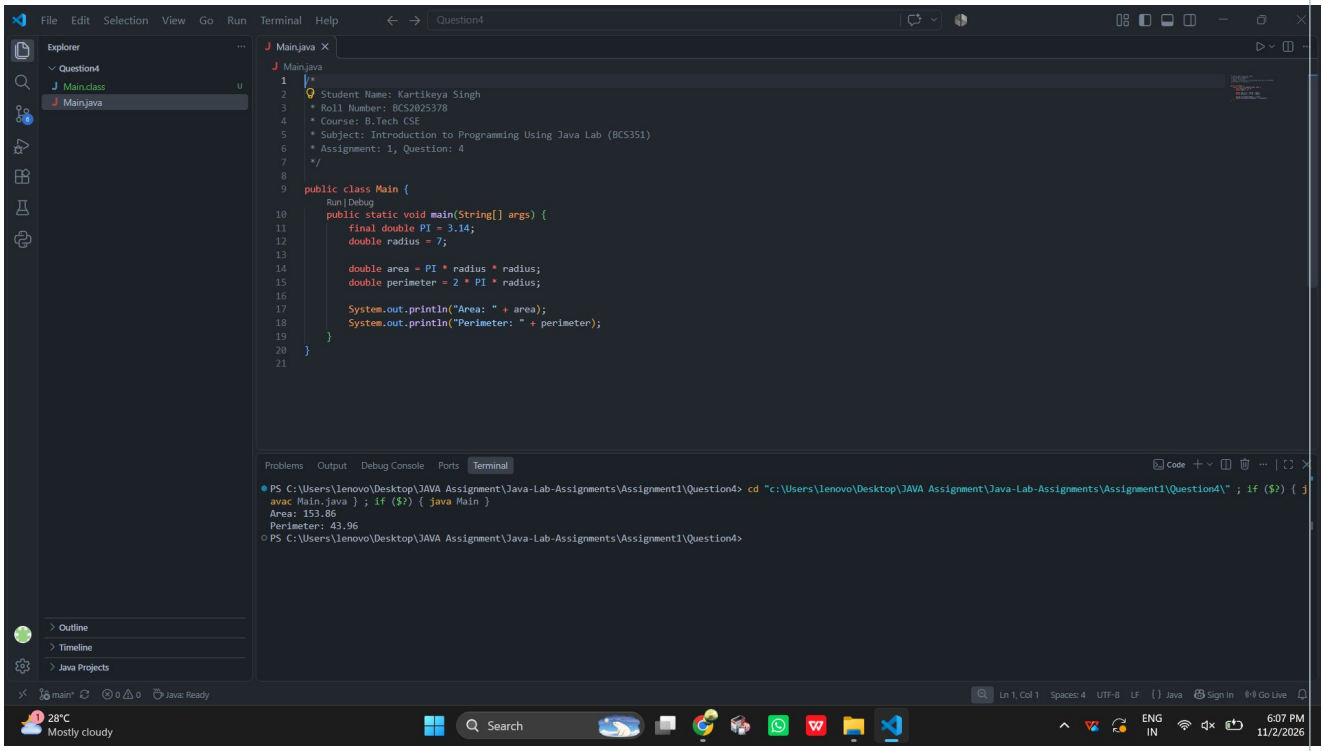
```
/*
 * Student Name: Kartikeya Singh
 * Roll Number: BCS2025378
 * Course: B.Tech CSE
 * Subject: Introduction to Programming Using Java Lab (BCS351)
 * Assignment: 1, Question: 4
 */

public class Main {
    public static void main(String[] args) {
        final double PI = 3.14;
        double radius = 7;

        double area = PI * radius * radius;
        double perimeter = 2 * PI * radius;
```

```
        System.out.println("Area: " + area);
        System.out.println("Perimeter: " + perimeter);
    }
}
```

Execution Output Screenshot:



Assignment 2: Decisions and Loops with Numbers

Assignment Aim / Objective: Use if, else if and loops to make decisions and repeat steps with whole numbers.

Question 1

Problem Statement: Store 27 in an int named n. Print the value of n followed by is even or is odd, using the % operator and an if-else statement.

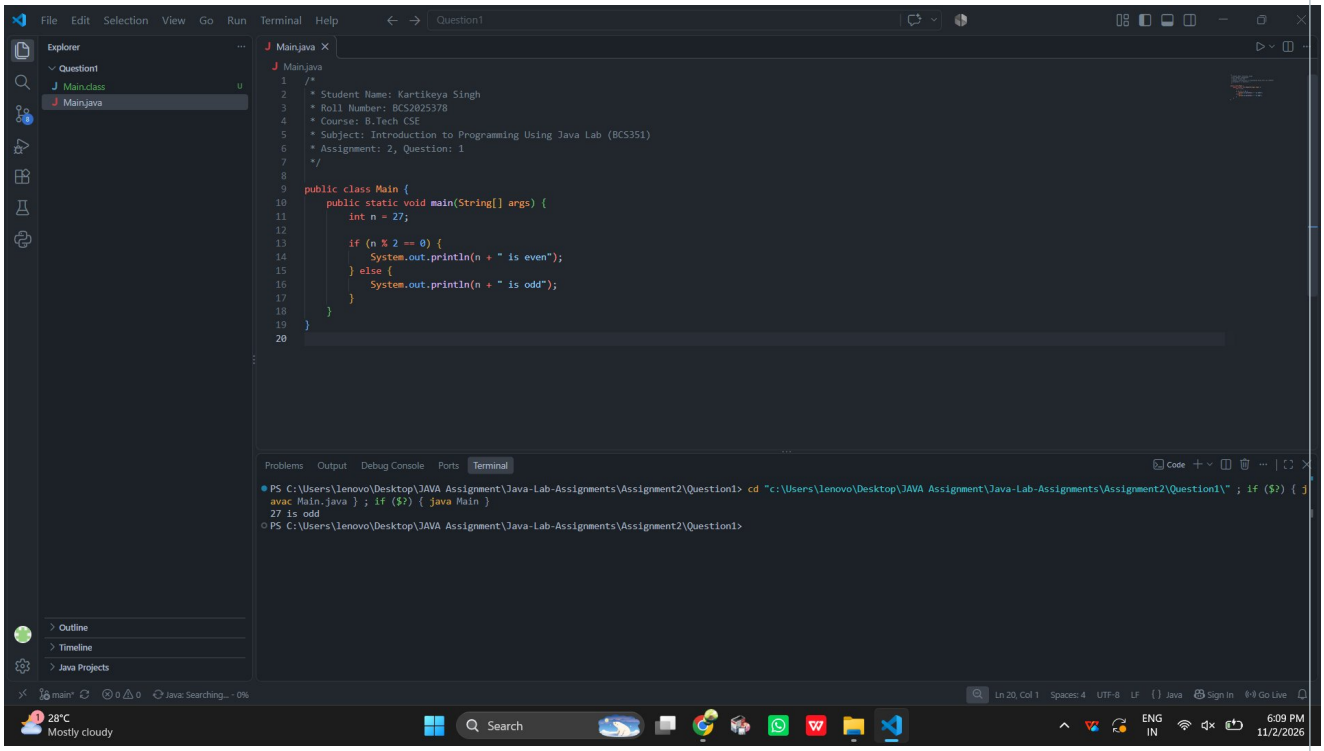
Source Code (Main.java):

```
/*
 * Student Name: Kartikeya Singh
 * Roll Number: BCS2025378
 * Course: B.Tech CSE
 * Subject: Introduction to Programming Using Java Lab (BCS351)
 * Assignment: 2, Question: 1
 */

public class Main {
    public static void main(String[] args) {
        int n = 27;

        if (n % 2 == 0) {
            System.out.println(n + " is even");
        } else {
            System.out.println(n + " is odd");
        }
    }
}
```

Execution Output Screenshot:



Question 2

Problem Statement: Store 83 in an int named marks. Print Grade: followed by A for 90 or more, B for 75 to 89, and C for less than 75. Use an else-if ladder.

Source Code (Main.java):

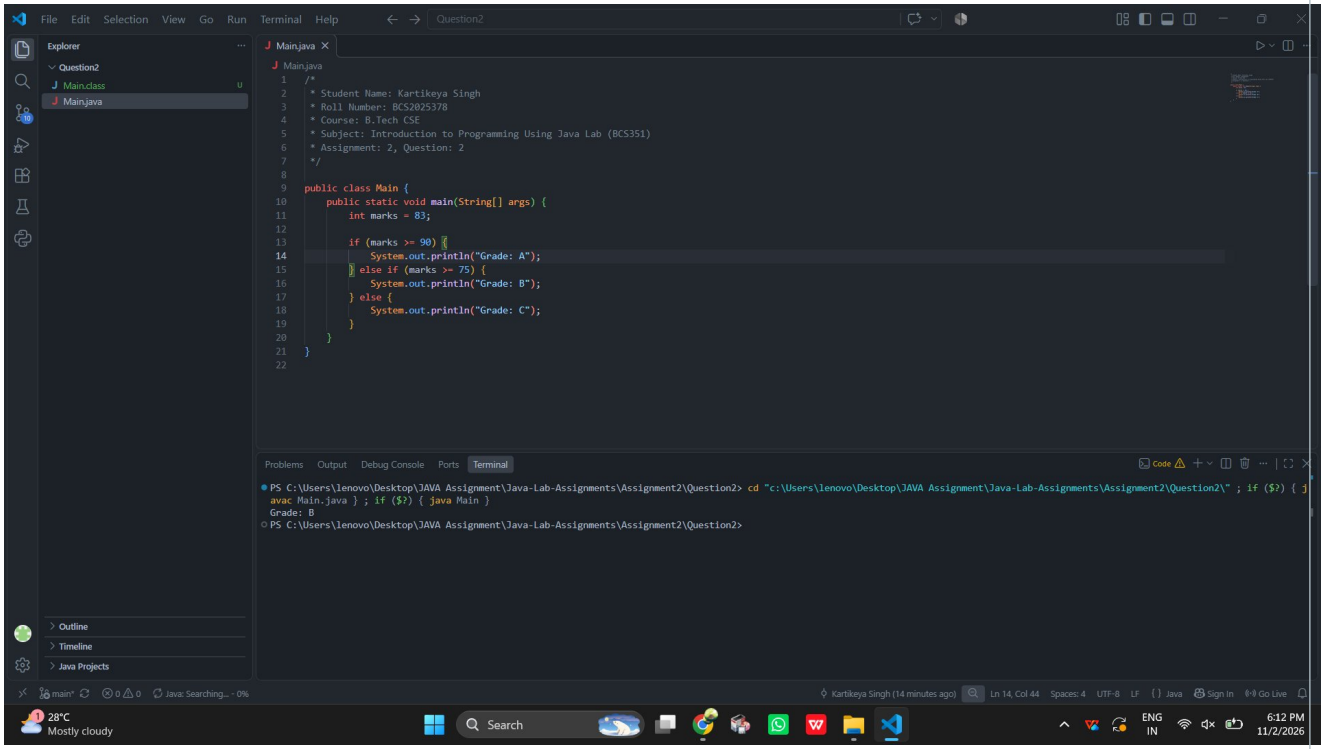
```
/*
 * Student Name: Kartikeya Singh
 * Roll Number: BCS2025378
 * Course: B.Tech CSE
 * Subject: Introduction to Programming Using Java Lab (BCS351)
 * Assignment: 2, Question: 2
 */

public class Main {
    public static void main(String[] args) {
        int marks = 83;

        if (marks >= 90) {
            System.out.println("Grade: A");
        } else if (marks >= 75) {
            System.out.println("Grade: B");
        } else {
            System.out.println("Grade: C");
        }
```

```
    }
}
```

Execution Output Screenshot:



Question 3

Problem Statement: Store 2024 in an int named year. A leap year is divisible by 4 but not by 100, or divisible by 400. Print Leap year for a leap year, otherwise Not a leap year.

Source Code (Main.java):

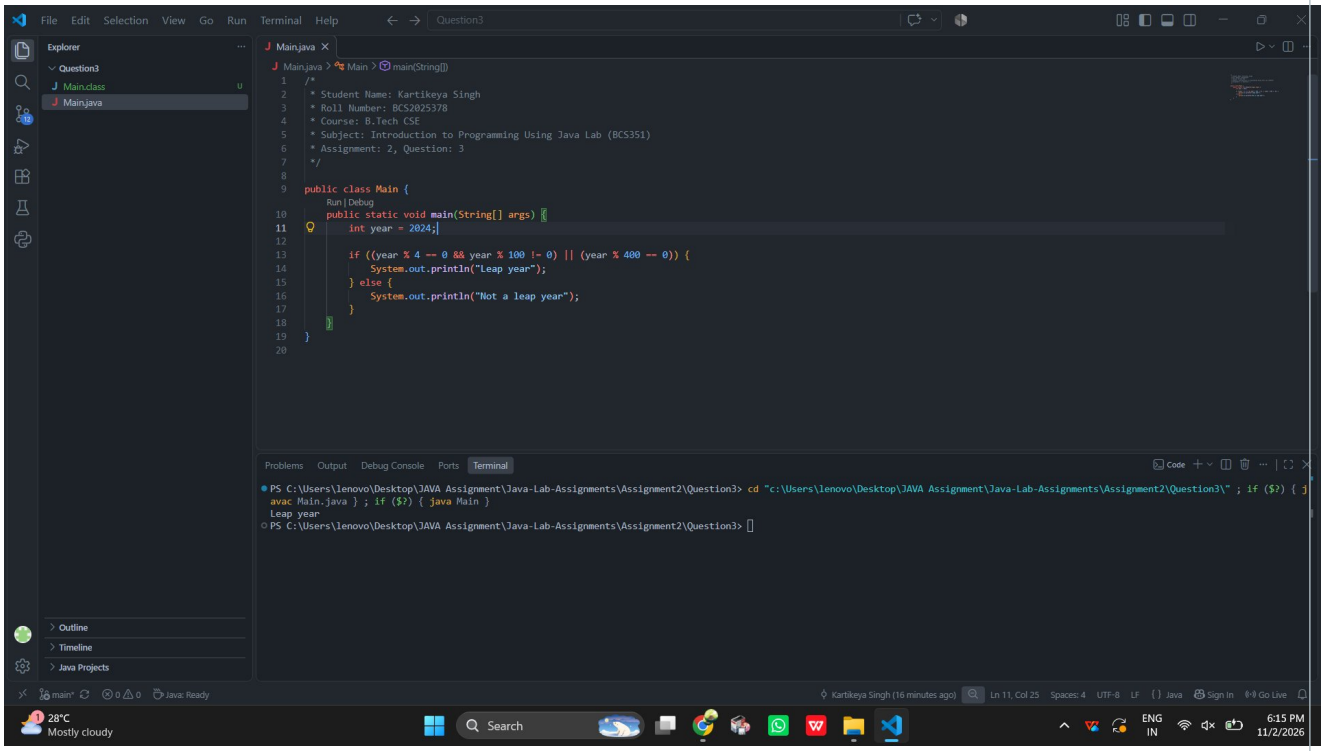
```
/*
 * Student Name: Kartikeya Singh
 * Roll Number: BCS2025378
 * Course: B.Tech CSE
 * Subject: Introduction to Programming Using Java Lab (BCS351)
 * Assignment: 2, Question: 3
 */

public class Main {
    public static void main(String[] args) {
        int year = 2024;

        if ((year % 4 == 0 && year % 100 != 0) || (year % 400 == 0)) {
            System.out.println("Leap year");
        } else {
```

```
            System.out.println("Not a leap year");
        }
    }
}
```

Execution Output Screenshot:



Question 4

Problem Statement: Use a for loop to find the factorial of 5. Print Factorial of 5 is followed by the value.

Source Code (Main.java):

```
/*
 * Student Name: Kartikeya Singh
 * Roll Number: BCS2025378
 * Course: B.Tech CSE
 * Subject: Introduction to Programming Using Java Lab (BCS351)
 * Assignment: 2, Question: 4
 */

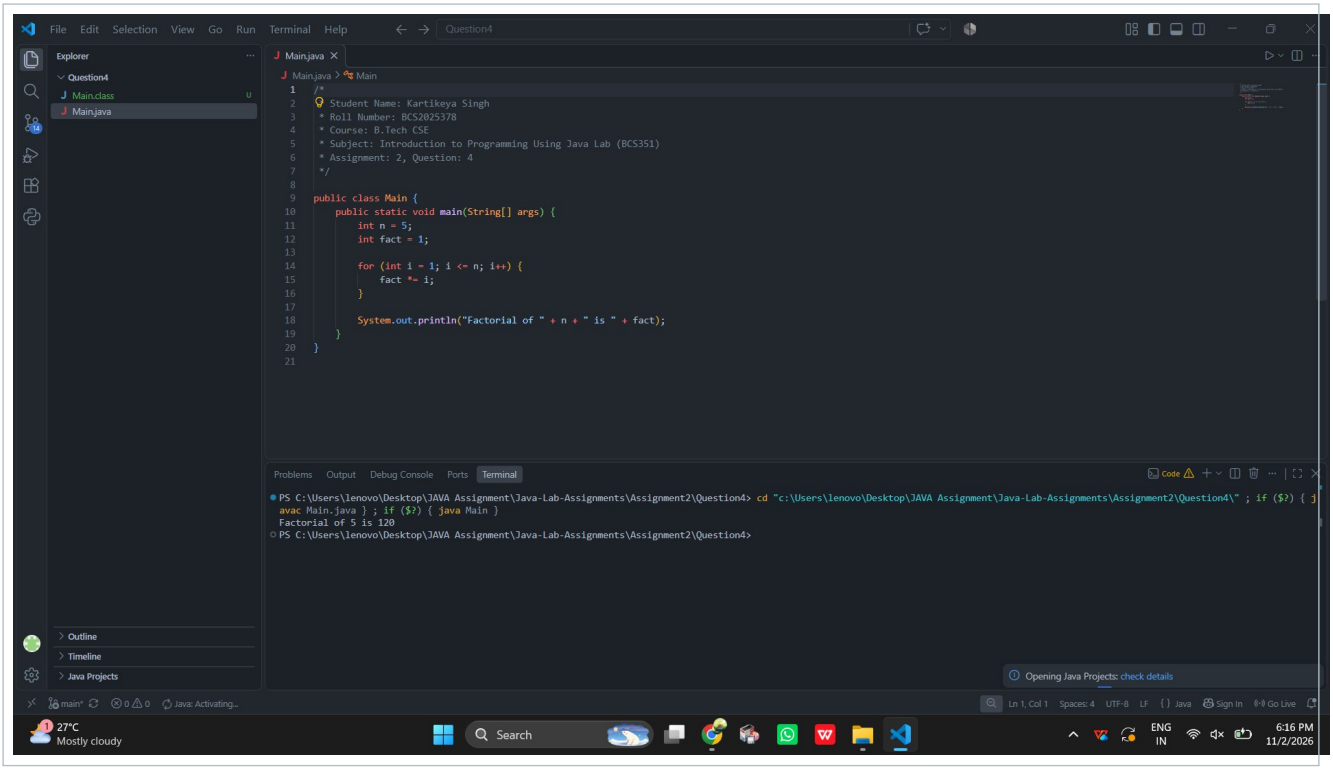
public class Main {
    public static void main(String[] args) {
        int n = 5;
        int fact = 1;

        for (int i = 1; i <= n; i++) {
```

```
            fact *= i;
        }

        System.out.println("Factorial of " + n + " is " + fact);
    }
}
```

Execution Output Screenshot:



Assignment 3: Star and Number Patterns

Assignment Aim / Objective: Use a loop inside another loop to print stars and numbers in rows.

Question 1

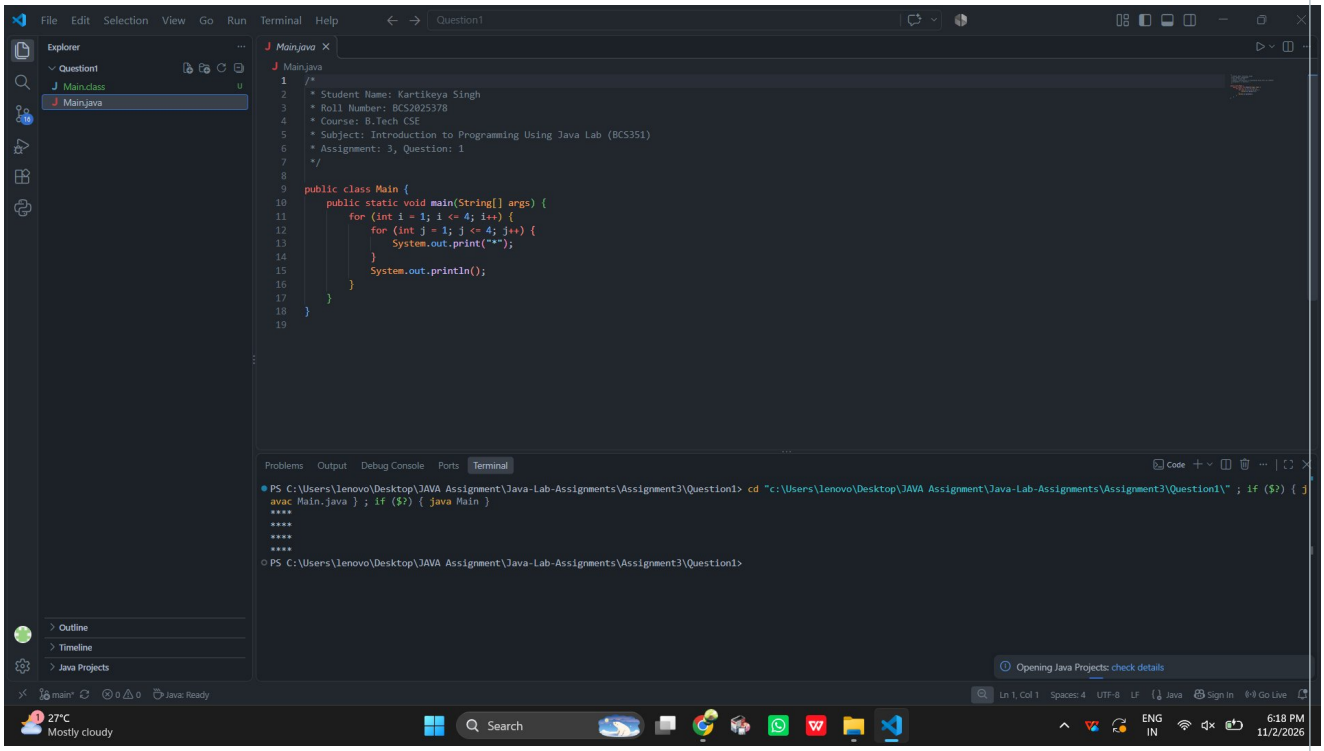
Problem Statement: Print a square of stars with 4 rows and 4 columns. Use one loop inside another loop.

Source Code (Main.java):

```
/*
 * Student Name: Kartikeya Singh
 * Roll Number: BCS2025378
 * Course: B.Tech CSE
 * Subject: Introduction to Programming Using Java Lab (BCS351)
 * Assignment: 3, Question: 1
 */

public class Main {
    public static void main(String[] args) {
        for (int i = 1; i <= 4; i++) {
            for (int j = 1; j <= 4; j++) {
                System.out.print("*");
            }
            System.out.println();
        }
    }
}
```

Execution Output Screenshot:



Question 2

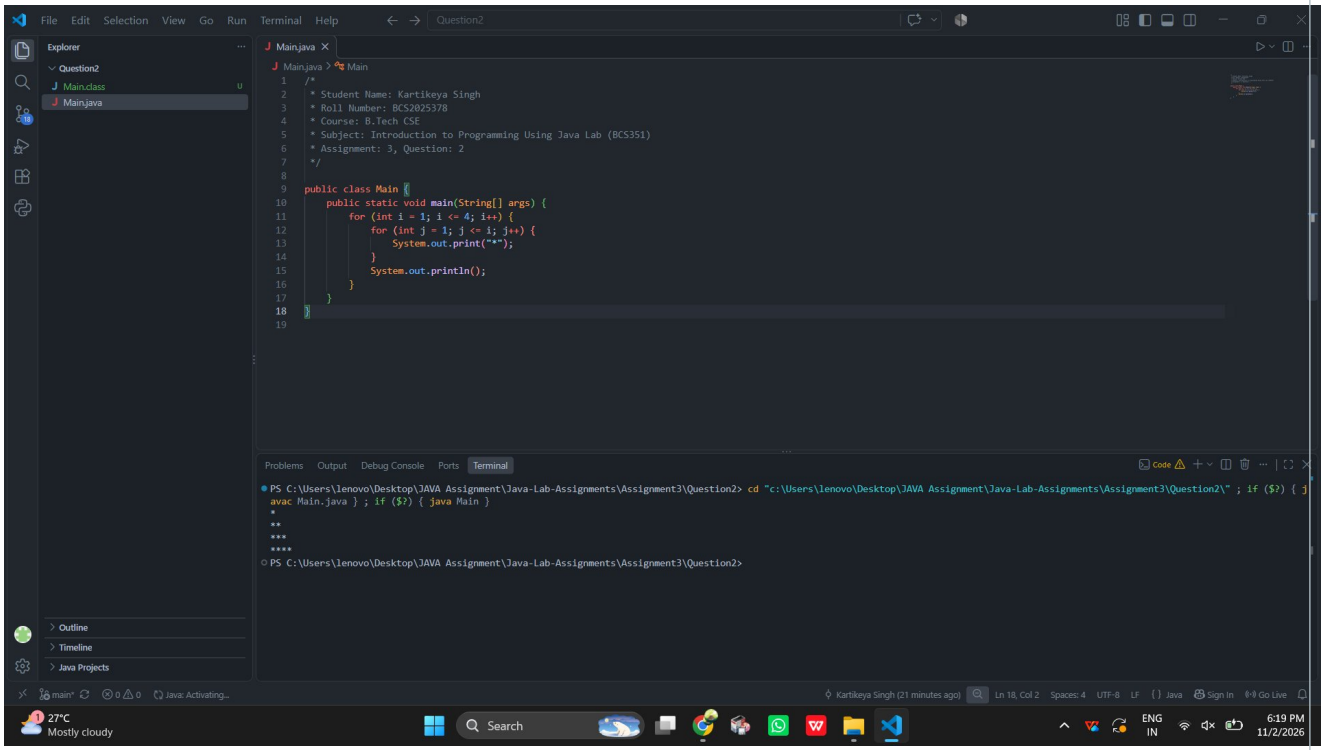
Problem Statement: Print a right triangle of stars with 4 rows. Row 1 has one star, row 2 has two stars, and so on.

Source Code (Main.java):

```
/*
 * Student Name: Kartikeya Singh
 * Roll Number: BCS2025378
 * Course: B.Tech CSE
 * Subject: Introduction to Programming Using Java Lab (BCS351)
 * Assignment: 3, Question: 2
 */

public class Main {
    public static void main(String[] args) {
        for (int i = 1; i <= 4; i++) {
            for (int j = 1; j <= i; j++) {
                System.out.print("*");
            }
            System.out.println();
        }
    }
}
```

Execution Output Screenshot:



Question 3

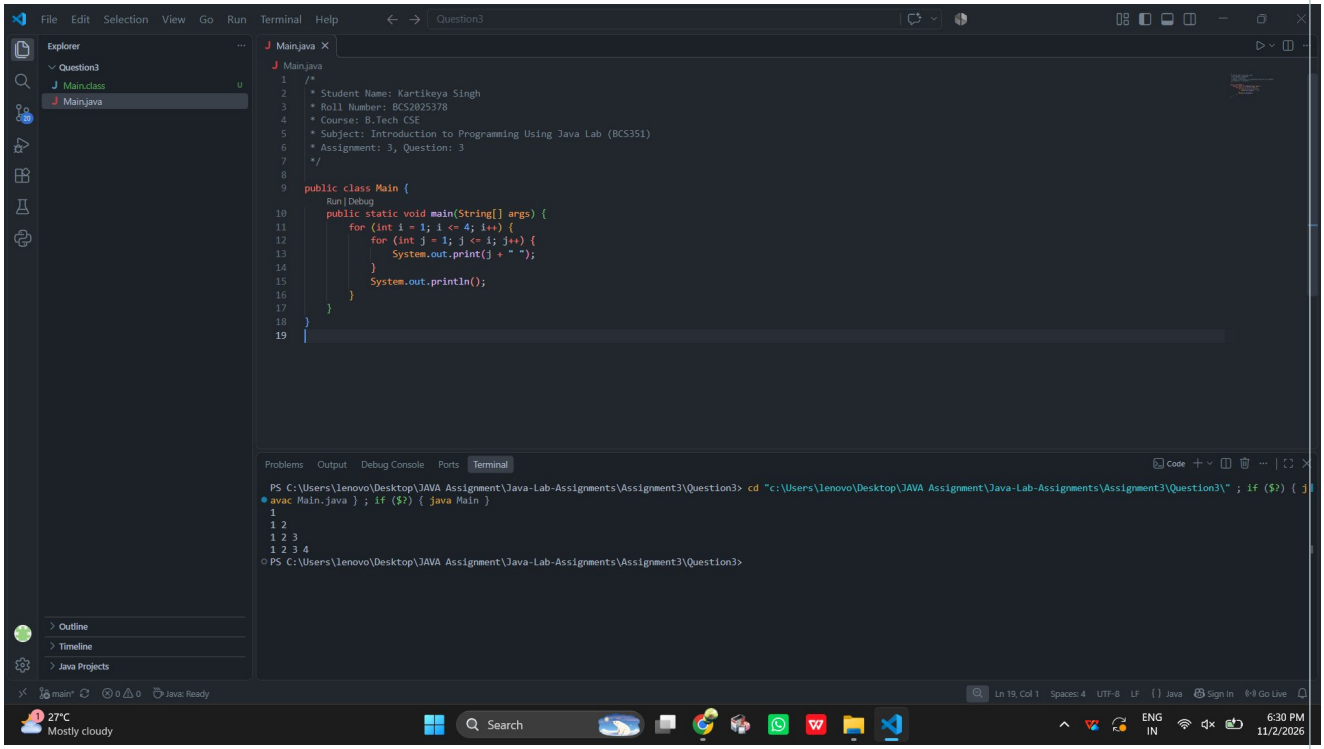
Problem Statement: Print a number triangle with 4 rows. Row 1 prints 1. Row 2 prints 1 2. Row 3 prints 1 2 3, and so on. Put a space after each number.

Source Code (Main.java):

```
/*
 * Student Name: Kartikeya Singh
 * Roll Number: BCS2025378
 * Course: B.Tech CSE
 * Subject: Introduction to Programming Using Java Lab (BCS351)
 * Assignment: 3, Question: 3
 */

public class Main {
    public static void main(String[] args) {
        for (int i = 1; i <= 4; i++) {
            for (int j = 1; j <= i; j++) {
                System.out.print(j + " ");
            }
            System.out.println();
        }
    }
}
```

Execution Output Screenshot:



Question 4

Problem Statement: Print an upside-down triangle of stars with 4 rows. Row 1 has four stars, row 2 has three stars, and so on, down to one star in row 4.

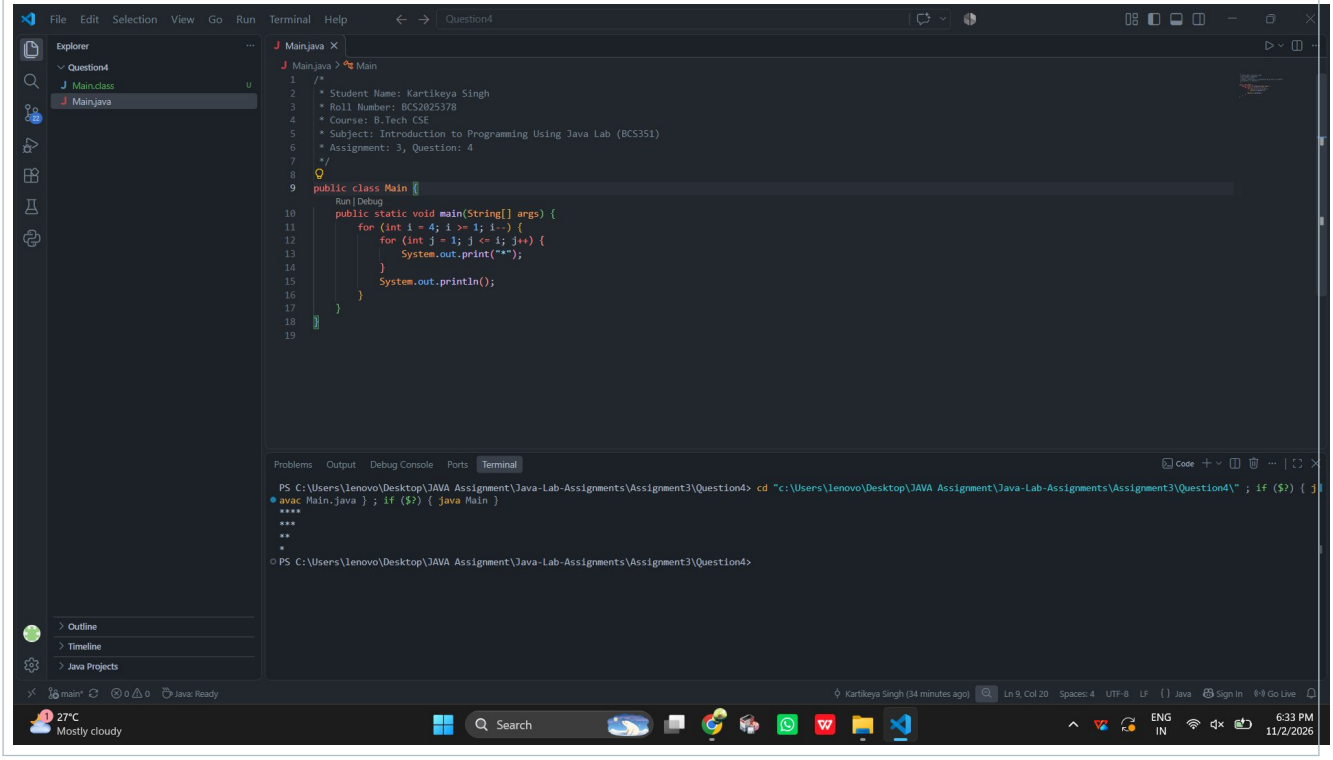
Source Code (Main.java):

```
/*
 * Student Name: Kartikeya Singh
 * Roll Number: BCS2025378
 * Course: B.Tech CSE
 * Subject: Introduction to Programming Using Java Lab (BCS351)
 * Assignment: 3, Question: 4
 */

public class Main {
    public static void main(String[] args) {
        for (int i = 4; i >= 1; i--) {
            for (int j = 1; j <= i; j++) {
                System.out.print("*");
            }
            System.out.println();
        }
    }
```

```
}
```

Execution Output Screenshot:



Assignment 4: Writing Functions

Assignment Aim / Objective: Write functions with parameters and return values, and call them from main.

Question 1

Problem Statement: Write a function fact(int n) that returns n factorial, using a loop, and a function nCr(int n, int r) that returns n C r, using fact. In main, print 5C2 = followed by the result of nCr(5, 2).

Source Code (Main.java):

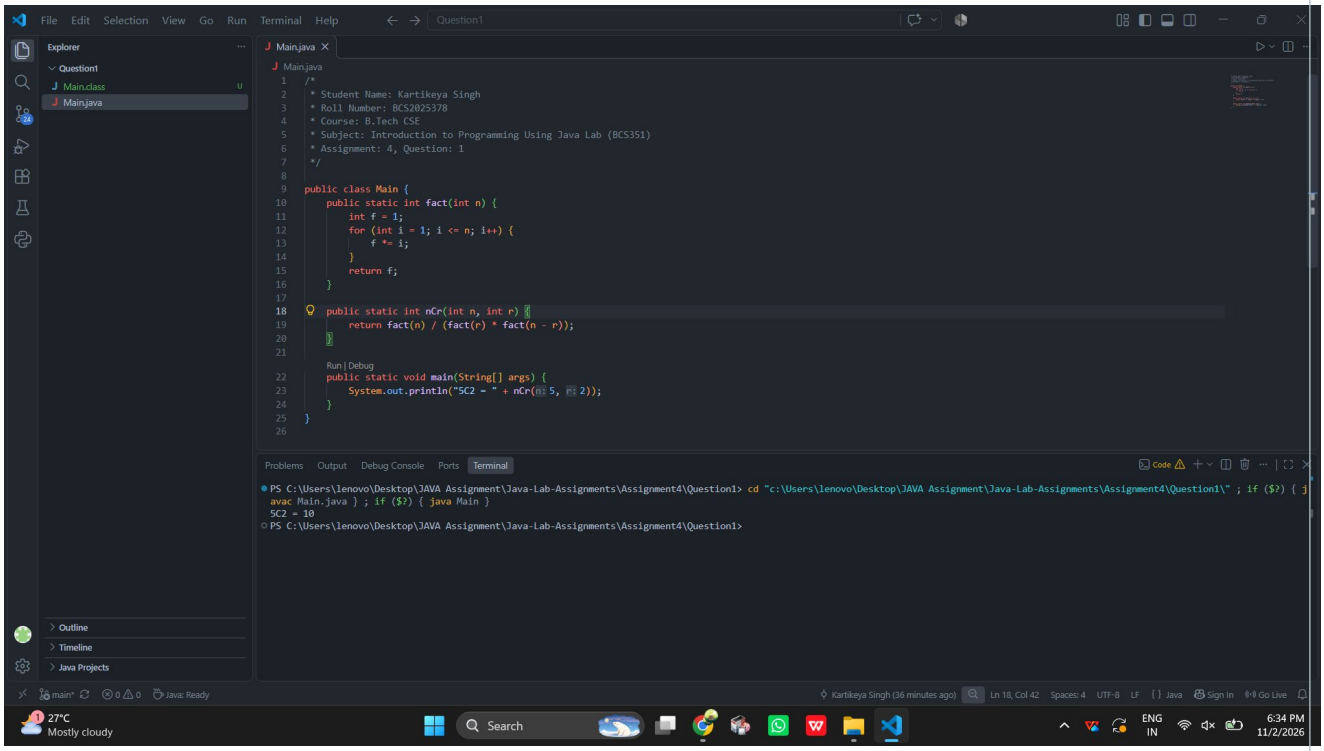
```
/*
 * Student Name: Kartikeya Singh
 * Roll Number: BCS2025378
 * Course: B.Tech CSE
 * Subject: Introduction to Programming Using Java Lab (BCS351)
 * Assignment: 4, Question: 1
 */

public class Main {
    public static int fact(int n) {
        int f = 1;
        for (int i = 1; i <= n; i++) {
            f *= i;
        }
        return f;
    }

    public static int nCr(int n, int r) {
        return fact(n) / (fact(r) * fact(n - r));
    }

    public static void main(String[] args) {
        System.out.println("5C2 = " + nCr(5, 2));
    }
}
```

Execution Output Screenshot:



Question 2

Problem Statement: Write fact(int n) that returns n factorial using a loop, and nPr(int n, int r) that returns n P r using fact. In main, print 5P2 = followed by the result of nPr(5, 2).

Source Code (Main.java):

```
/*
 * Student Name: Kartikeya Singh
 * Roll Number: BCS2025378
 * Course: B.Tech CSE
 * Subject: Introduction to Programming Using Java Lab (BCS351)
 * Assignment: 4, Question: 2
 */

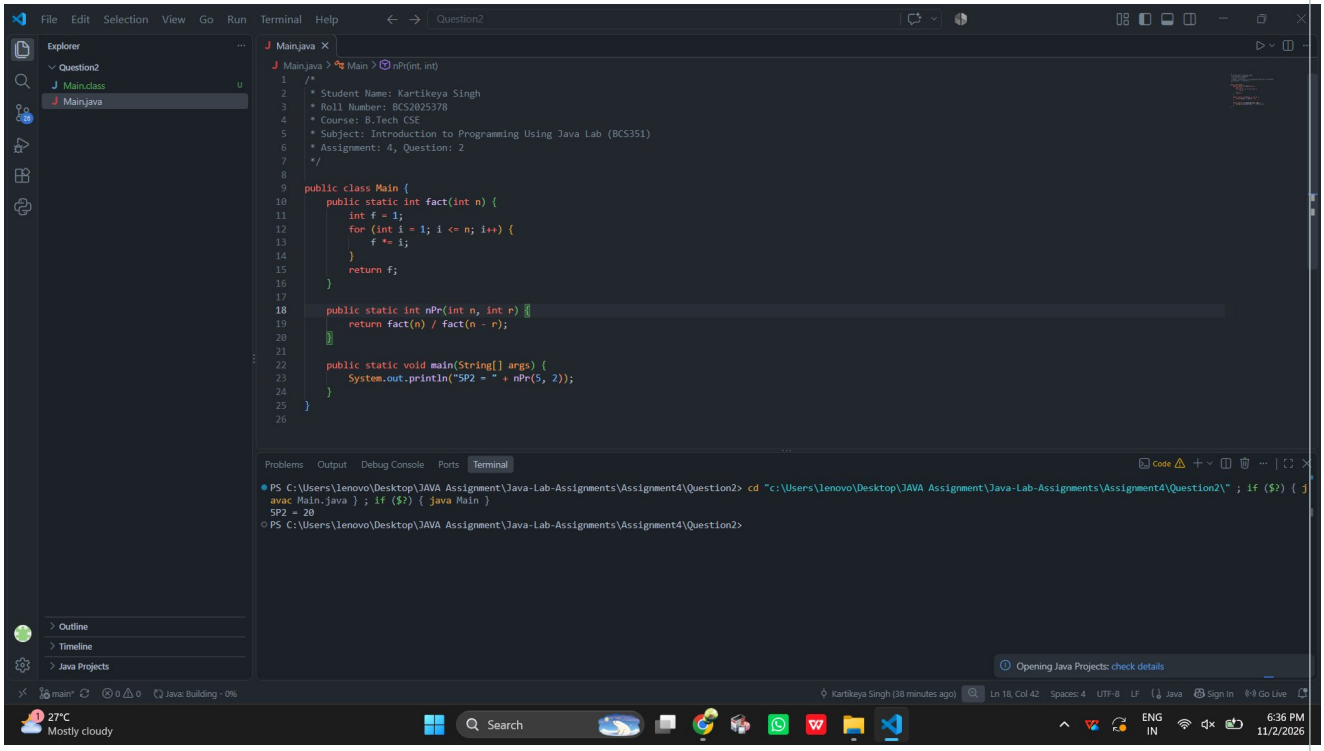
public class Main {
    public static int fact(int n) {
        int f = 1;
        for (int i = 1; i <= n; i++) {
            f *= i;
        }
        return f;
    }

    public static int nPr(int n, int r) {
        return fact(n) / fact(n - r);
```

```
    }

    public static void main(String[] args) {
        System.out.println("5P2 = " + nPr(5, 2));
    }
}
```

Execution Output Screenshot:



Question 3

Problem Statement: Write maxOf(int a, int b) that returns the larger of its two int values. In main, print Larger: followed by the result of maxOf(14, 9).

Source Code (Main.java):

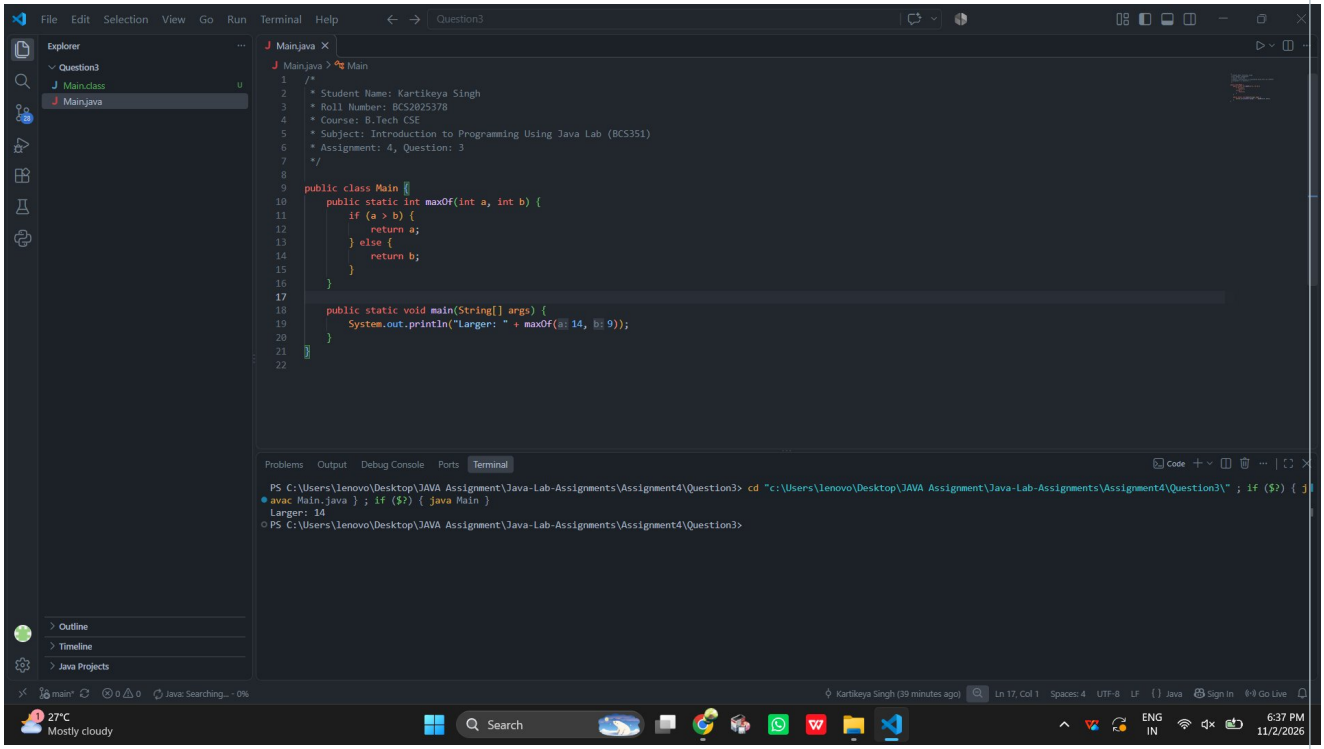
```
/*
 * Student Name: Kartikeya Singh
 * Roll Number: BCS2025378
 * Course: B.Tech CSE
 * Subject: Introduction to Programming Using Java Lab (BCS351)
 * Assignment: 4, Question: 3
 */

public class Main {
    public static int maxOf(int a, int b) {
        if (a > b) {
```

```
            return a;
        } else {
            return b;
        }
    }

    public static void main(String[] args) {
        System.out.println("Larger: " + maxOf(14, 9));
    }
}
```

Execution Output Screenshot:



Assignment 5: Arrays: Traversal and Counting

Assignment Aim / Objective: Create an array, visit every element with a loop, and add up or count the values.

Question 1

Problem Statement: Create an int array named arr with the values 3, 4 and 5. Use a for loop with arr.length to print one line for each element: the label Index, the index number, a colon and the element value.

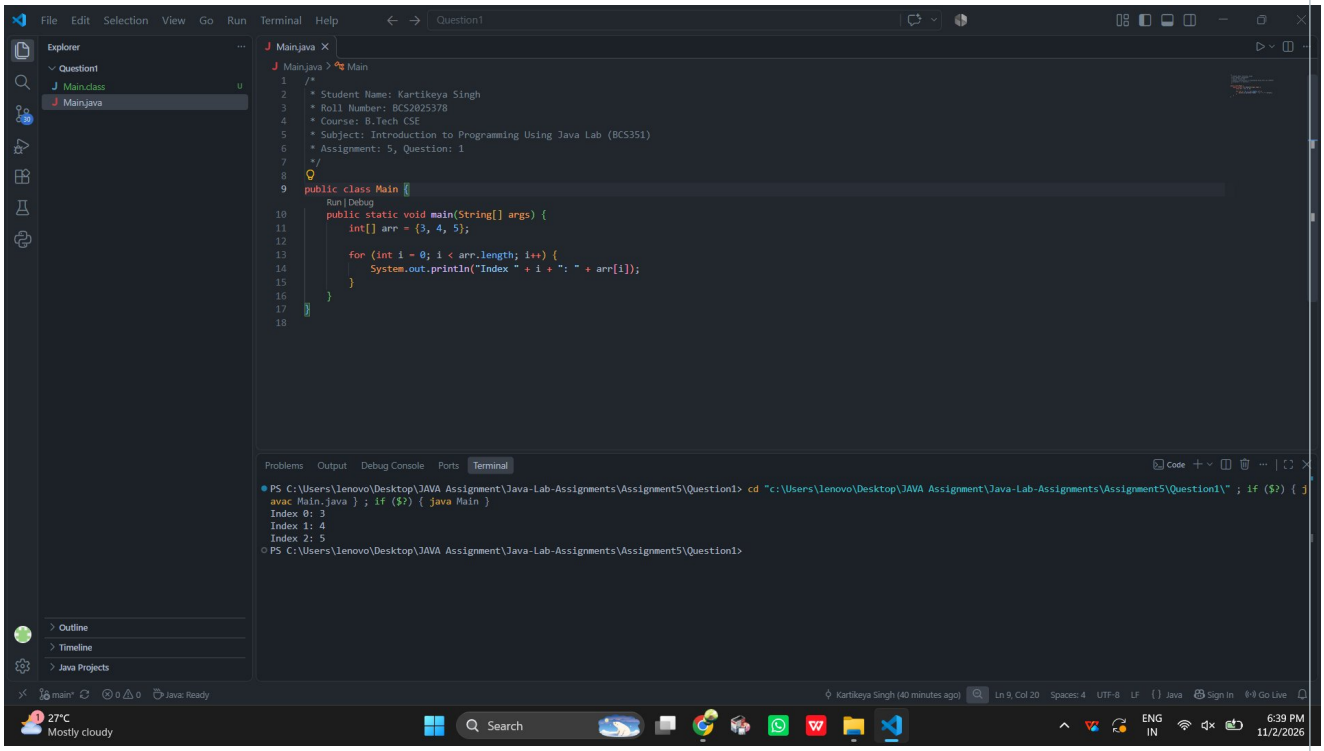
Source Code (Main.java):

```
/*
 * Student Name: Kartikeya Singh
 * Roll Number: BCS2025378
 * Course: B.Tech CSE
 * Subject: Introduction to Programming Using Java Lab (BCS351)
 * Assignment: 5, Question: 1
 */

public class Main {
    public static void main(String[] args) {
        int[] arr = {3, 4, 5};

        for (int i = 0; i < arr.length; i++) {
            System.out.println("Index " + i + ": " + arr[i]);
        }
    }
}
```

Execution Output Screenshot:



Question 2

Problem Statement: Create an int array with the values 12, 45, 7, 33 and 21. Use a loop to find the sum and the largest value. Print Sum: followed by the sum, then Largest: followed by the largest value.

Source Code (Main.java):

```
/*
 * Student Name: Kartikeya Singh
 * Roll Number: BCS2025378
 * Course: B.Tech CSE
 * Subject: Introduction to Programming Using Java Lab (BCS351)
 * Assignment: 5, Question: 2
 */

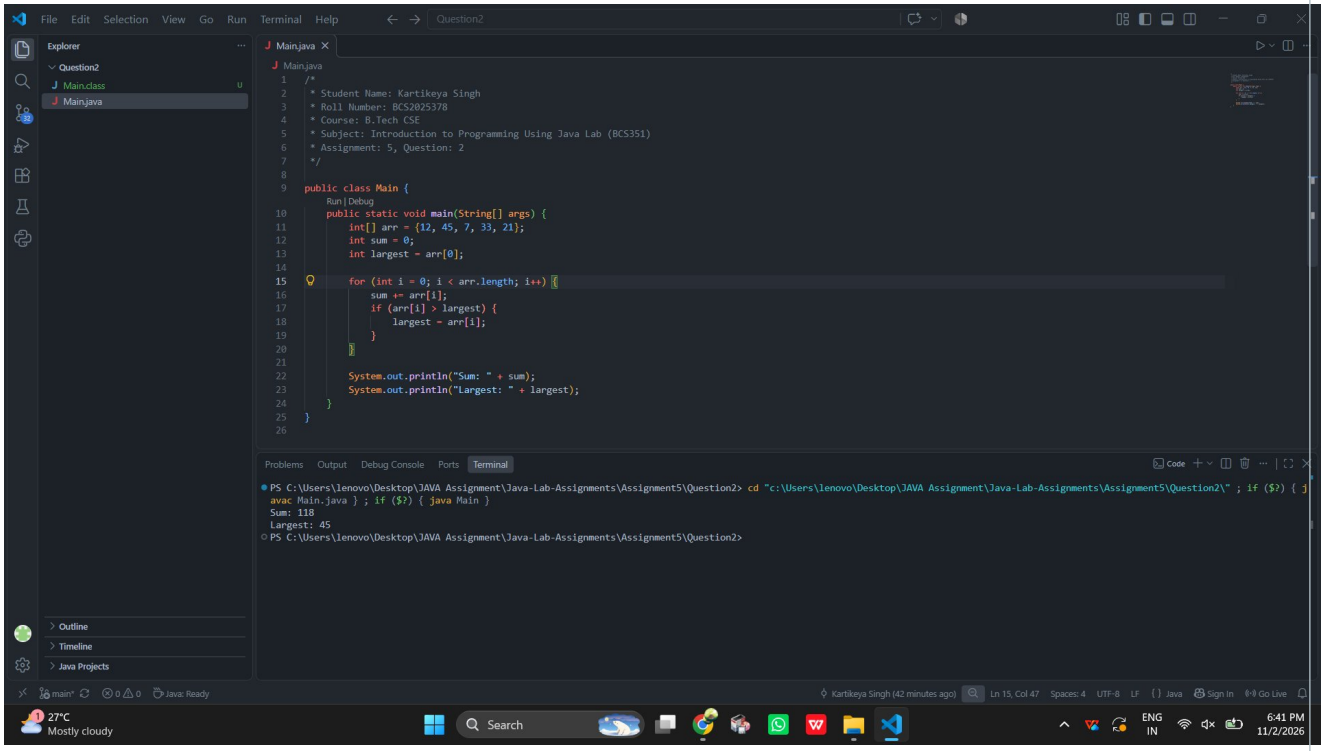
public class Main {
    public static void main(String[] args) {
        int[] arr = {12, 45, 7, 33, 21};
        int sum = 0;
        int largest = arr[0];

        for (int i = 0; i < arr.length; i++) {
            sum += arr[i];
            if (arr[i] > largest) {
                largest = arr[i];
            }
```

```
        }

        System.out.println("Sum: " + sum);
        System.out.println("Largest: " + largest);
    }
}
```

Execution Output Screenshot:



Question 3

Problem Statement: Create an int array with the values 5, 12, 2, 14 and 7. Use a loop to count the values greater than 10. Print Count greater than 10: followed by the count.

Source Code (Main.java):

```
/*
 * Student Name: Kartikeya Singh
 * Roll Number: BCS2025378
 * Course: B.Tech CSE
 * Subject: Introduction to Programming Using Java Lab (BCS351)
 * Assignment: 5, Question: 3
 */

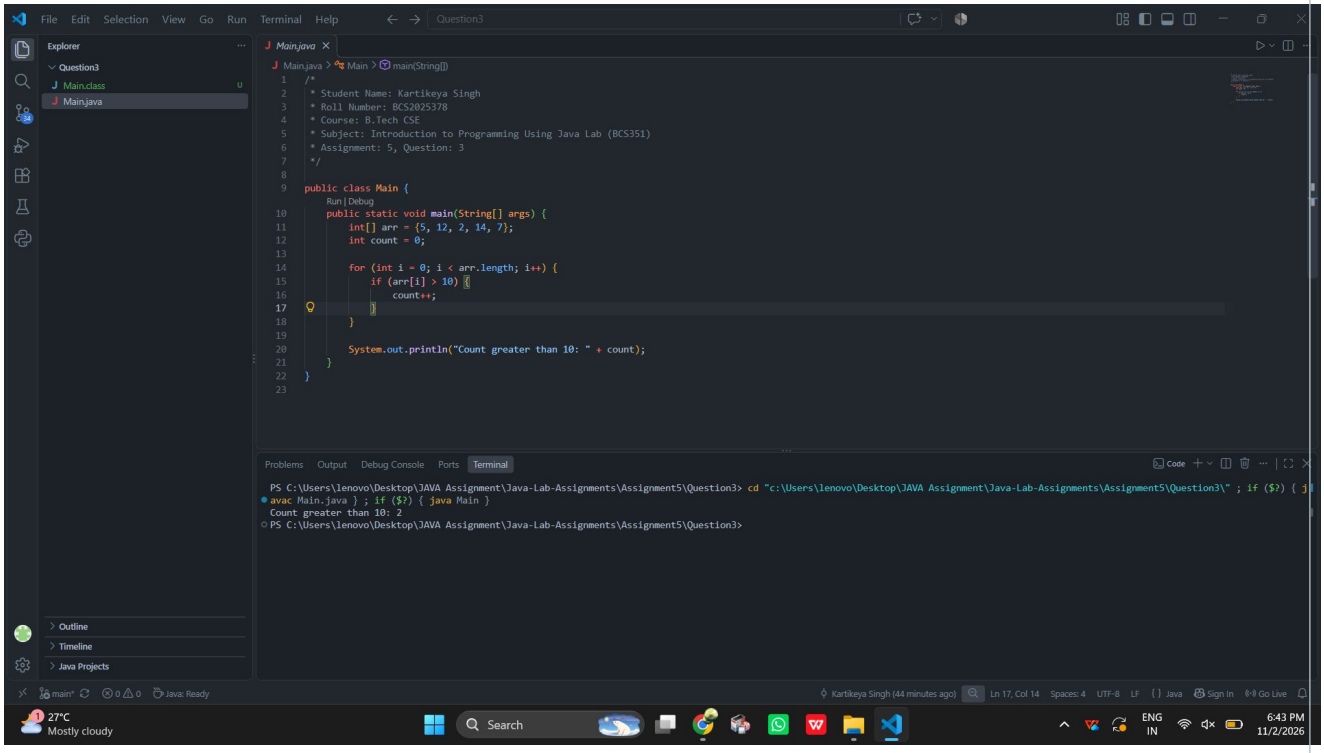
public class Main {
    public static void main(String[] args) {
        int[] arr = {5, 12, 2, 14, 7};
```

```
        int count = 0;

        for (int i = 0; i < arr.length; i++) {
            if (arr[i] > 10) {
                count++;
            }
        }

        System.out.println("Count greater than 10: " + count);
    }
}
```

Execution Output Screenshot:



Assignment 6: Arrays: Reverse and Duplicates

Assignment Aim / Objective: Print an array in reverse order, find the smallest value, and check an array for repeated values.

Question 1

Problem Statement: Create an int array with the values 10, 20, 30, 40 and 50. Print the values in reverse order, one value on each line, starting with the last value.

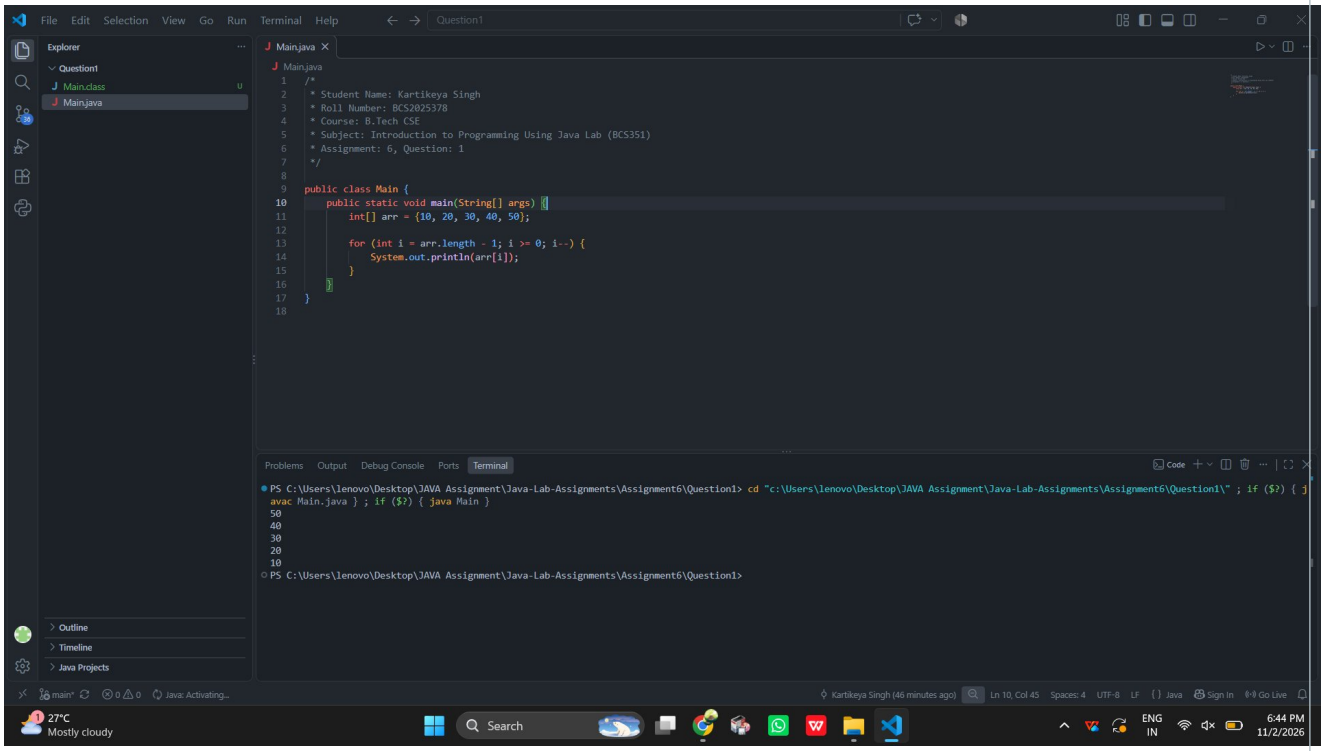
Source Code (Main.java):

```
/*
 * Student Name: Kartikeya Singh
 * Roll Number: BCS2025378
 * Course: B.Tech CSE
 * Subject: Introduction to Programming Using Java Lab (BCS351)
 * Assignment: 6, Question: 1
 */

public class Main {
    public static void main(String[] args) {
        int[] arr = {10, 20, 30, 40, 50};

        for (int i = arr.length - 1; i >= 0; i--) {
            System.out.println(arr[i]);
        }
    }
}
```

Execution Output Screenshot:



Question 2

Problem Statement: Create an int array with the values 2, 5, 1, 8, 3 and 4. Use a loop to find the smallest value and its index. Print Smallest: followed by the value, then the words at index and the index.

Source Code (Main.java):

```
/*
 * Student Name: Kartikeya Singh
 * Roll Number: BCS2025378
 * Course: B.Tech CSE
 * Subject: Introduction to Programming Using Java Lab (BCS351)
 * Assignment: 6, Question: 2
 */

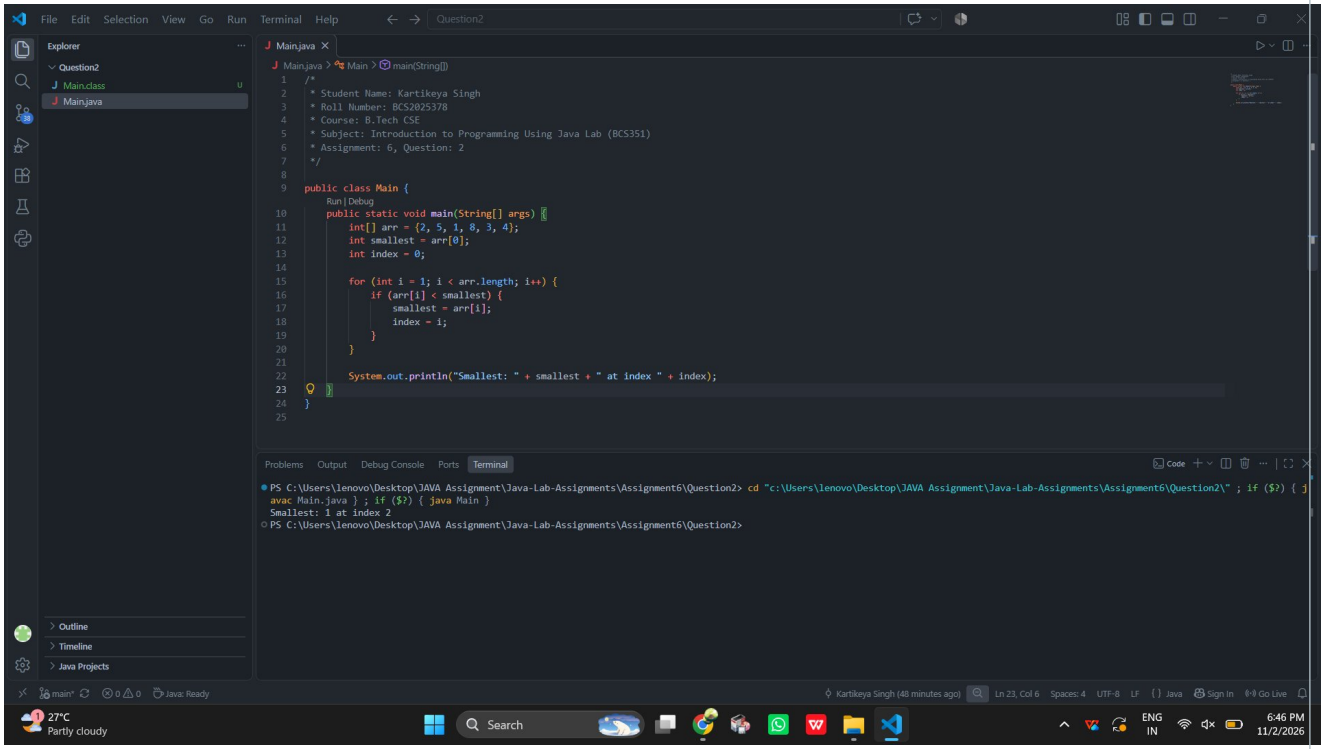
public class Main {
    public static void main(String[] args) {
        int[] arr = {2, 5, 1, 8, 3, 4};
        int smallest = arr[0];
        int index = 0;

        for (int i = 1; i < arr.length; i++) {
            if (arr[i] < smallest) {
                smallest = arr[i];
                index = i;
            }
```

```
        }

        System.out.println("Smallest: " + smallest + " at index " + index);
    }
}
```

Execution Output Screenshot:



Question 3

Problem Statement: Create an int array with the values 4, 9, 2, 9 and 7. Use nested loops to check for a repeated value. Print Repeated value: followed by it, or print No repeated value if none is repeated.

Source Code (Main.java):

```
/*
 * Student Name: Kartikeya Singh
 * Roll Number: BCS2025378
 * Course: B.Tech CSE
 * Subject: Introduction to Programming Using Java Lab (BCS351)
 * Assignment: 6, Question: 3
 */

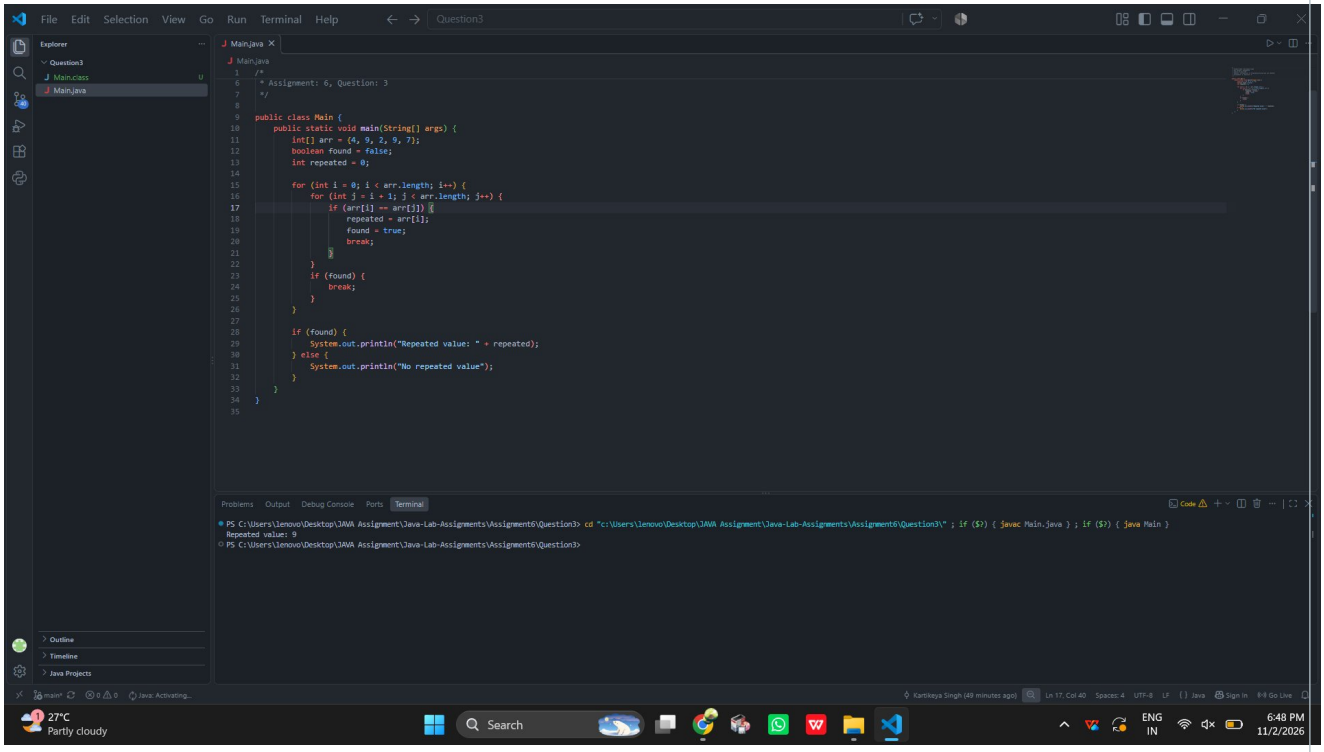
public class Main {
    public static void main(String[] args) {
        int[] arr = {4, 9, 2, 9, 7};
        boolean found = false;
```

```
        int repeated = 0;

        for (int i = 0; i < arr.length; i++) {
            for (int j = i + 1; j < arr.length; j++) {
                if (arr[i] == arr[j]) {
                    repeated = arr[i];
                    found = true;
                    break;
                }
            }
            if (found) {
                break;
            }
        }

        if (found) {
            System.out.println("Repeated value: " + repeated);
        } else {
            System.out.println("No repeated value");
        }
    }
}
```

Execution Output Screenshot:



Assignment 7: Counting Steps and Complexity

Assignment Aim / Objective: Count the steps a program takes, and write its time complexity as a comment.

Question 1

Problem Statement: Use one loop that runs from 1 to 10, and count its steps. Above the loop, write a comment with its time complexity. Print Steps: followed by the number of steps.

Source Code (Main.java):

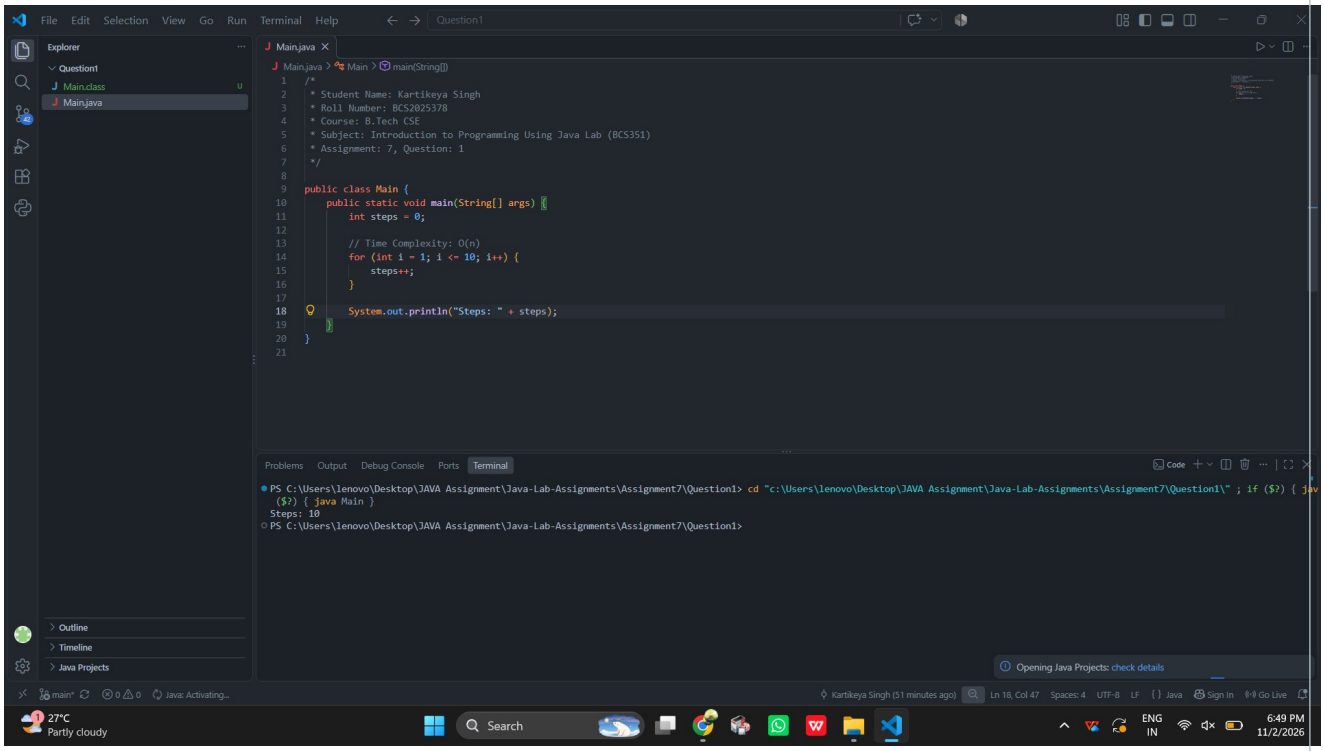
```
/*
 * Student Name: Kartikeya Singh
 * Roll Number: BCS2025378
 * Course: B.Tech CSE
 * Subject: Introduction to Programming Using Java Lab (BCS351)
 * Assignment: 7, Question: 1
 */

public class Main {
    public static void main(String[] args) {
        int steps = 0;

        // Time Complexity: O(n)
        for (int i = 1; i <= 10; i++) {
            steps++;
        }

        System.out.println("Steps: " + steps);
    }
}
```

Execution Output Screenshot:



Question 2

Problem Statement: Use two loops, one inside the other, that both run from 1 to 10. Count how many times the inner loop body runs. Above the loops, write a comment with their time complexity. Print Steps: followed by the total.

Source Code (Main.java):

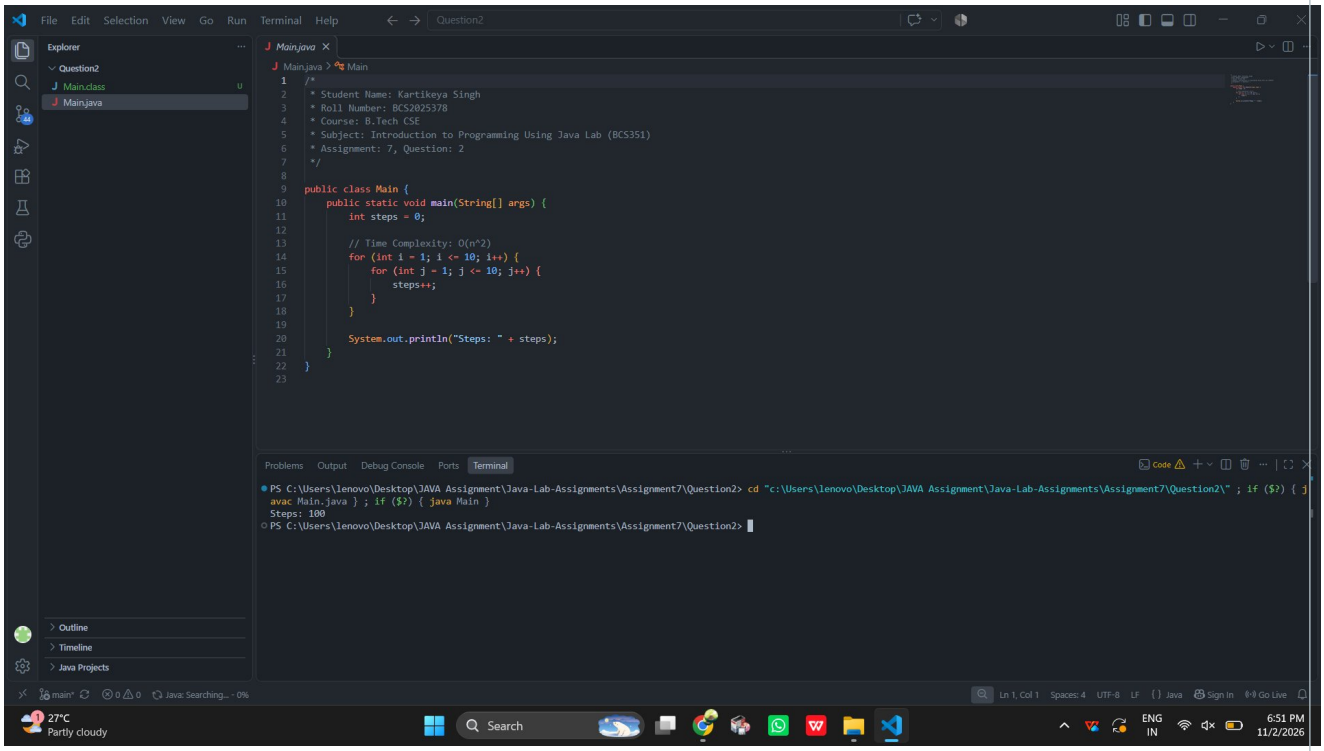
```
/*
 * Student Name: Kartikeya Singh
 * Roll Number: BCS2025378
 * Course: B.Tech CSE
 * Subject: Introduction to Programming Using Java Lab (BCS351)
 * Assignment: 7, Question: 2
 */

public class Main {
    public static void main(String[] args) {
        int steps = 0;

        // Time Complexity: O(n^2)
        for (int i = 1; i <= 10; i++) {
            for (int j = 1; j <= 10; j++) {
                steps++;
            }
        }
```

```
        System.out.println("Steps: " + steps);
    }
}
```

Execution Output Screenshot:



Question 3

Problem Statement: Store 1024 in an int named n. Use a while loop that halves n until n is 1, and count the divisions. Above the loop, write a comment with its time complexity. Print Halving steps: followed by the count.

Source Code (Main.java):

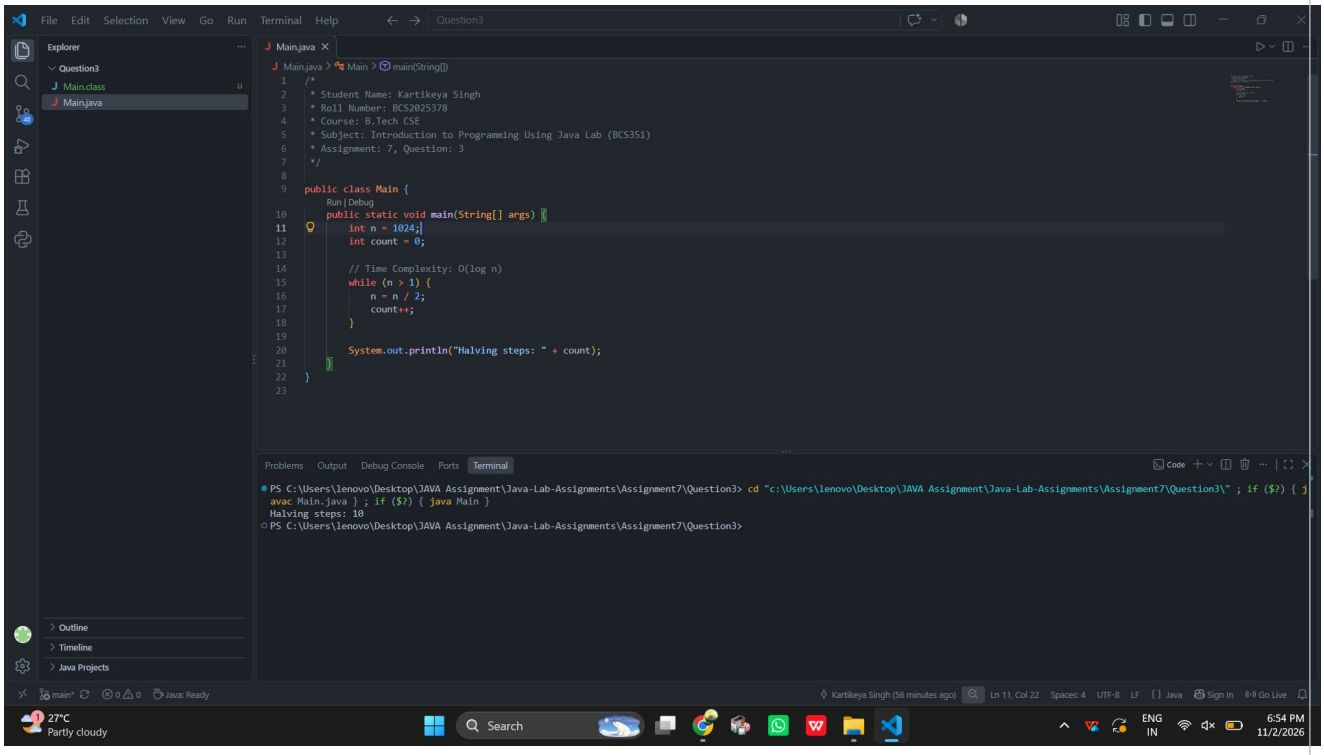
```
/*
 * Student Name: Kartikeya Singh
 * Roll Number: BCS2025378
 * Course: B.Tech CSE
 * Subject: Introduction to Programming Using Java Lab (BCS351)
 * Assignment: 7, Question: 3
 */

public class Main {
    public static void main(String[] args) {
        int n = 1024;
        int count = 0;
```

```
        // Time Complexity: O(log n)
        while (n > 1) {
            n = n / 2;
            count++;
        }

        System.out.println("Halving steps: " + count);
    }
}
```

Execution Output Screenshot:



Assignment 8: Matrices: Transpose and Row Sums

Assignment Aim / Objective: Create two-dimensional arrays, visit them row by row, and build a new matrix from one.

Question 1

Problem Statement: Create a matrix named m with 3 rows and 4 columns. Fill it with the values 1 to 12, row by row, using nested loops. Print the matrix one row per line, with a space after each number.

Source Code (Main.java):

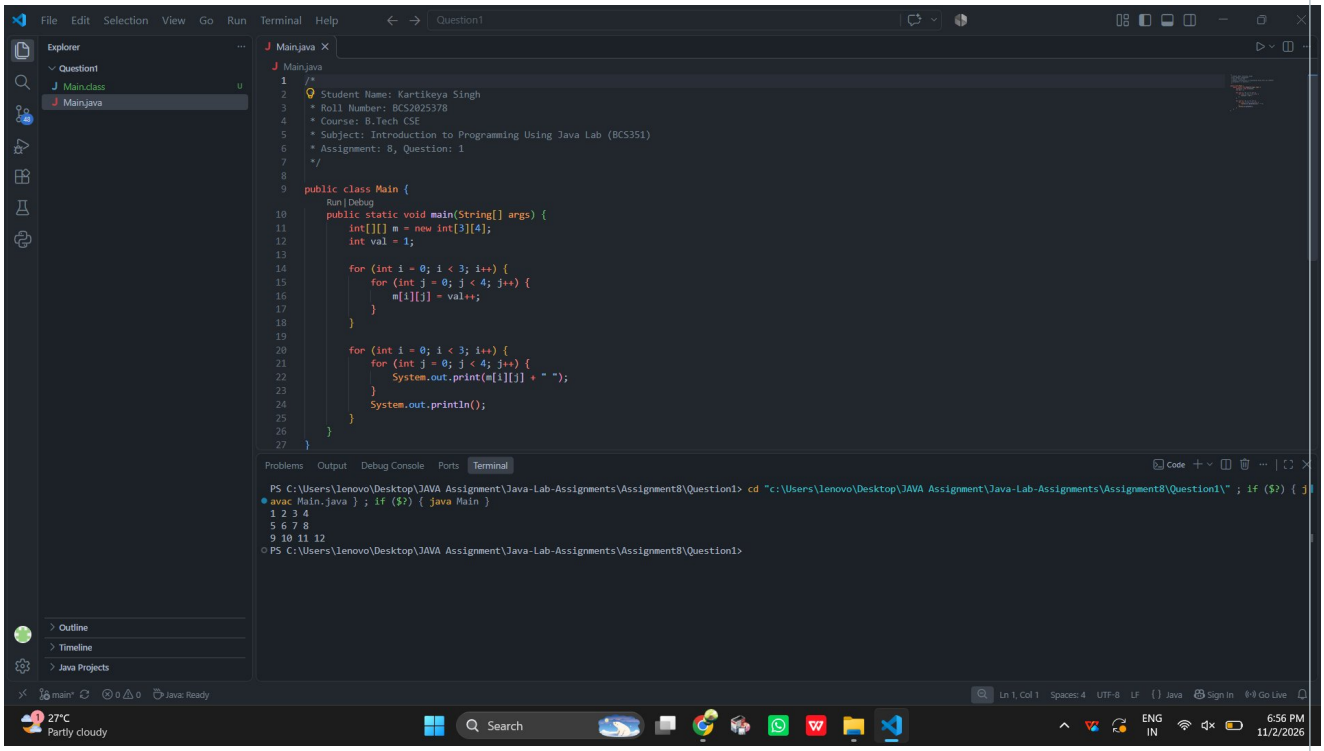
```
/*
 * Student Name: Kartikeya Singh
 * Roll Number: BCS2025378
 * Course: B.Tech CSE
 * Subject: Introduction to Programming Using Java Lab (BCS351)
 * Assignment: 8, Question: 1
 */

public class Main {
    public static void main(String[] args) {
        int[][] m = new int[3][4];
        int val = 1;

        for (int i = 0; i < 3; i++) {
            for (int j = 0; j < 4; j++) {
                m[i][j] = val++;
            }
        }

        for (int i = 0; i < 3; i++) {
            for (int j = 0; j < 4; j++) {
                System.out.print(m[i][j] + " ");
            }
            System.out.println();
        }
    }
}
```

Execution Output Screenshot:



Question 2

Problem Statement: Create the 3 by 3 matrix with the rows 1 2 3, 4 5 6 and 7 8 9. Create its transpose (rows and columns swapped). Print the transpose one row per line, with a space after each number.

Source Code (Main.java):

```
/*
 * Student Name: Kartikeya Singh
 * Roll Number: BCS2025378
 * Course: B.Tech CSE
 * Subject: Introduction to Programming Using Java Lab (BCS351)
 * Assignment: 8, Question: 2
 */

public class Main {
    public static void main(String[] args) {
        int[][] matrix = {
            {1, 2, 3},
            {4, 5, 6},
            {7, 8, 9}
        };

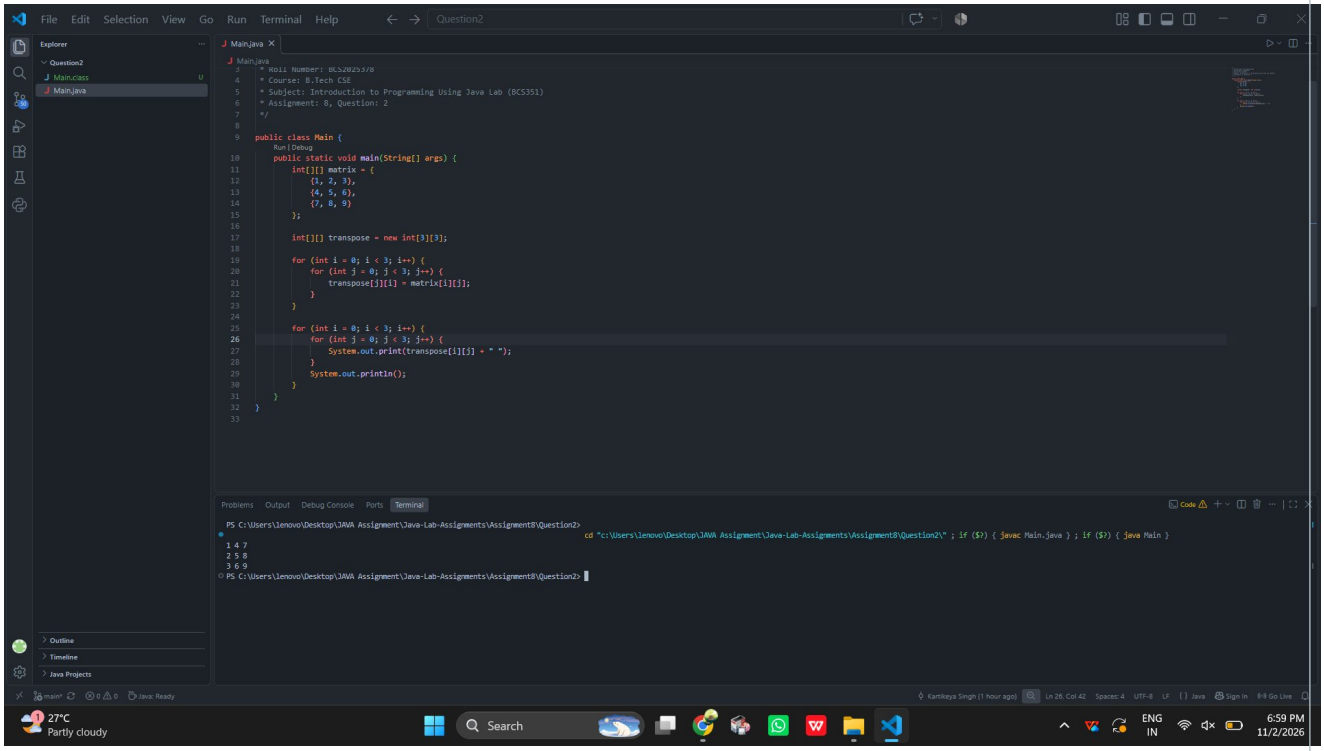
        int[][] transpose = new int[3][3];

        for (int i = 0; i < 3; i++) {
```

```
            for (int j = 0; j < 3; j++) {
                transpose[j][i] = matrix[i][j];
            }
        }

        for (int i = 0; i < 3; i++) {
            for (int j = 0; j < 3; j++) {
                System.out.print(transpose[i][j] + " ");
            }
            System.out.println();
        }
    }
}
```

Execution Output Screenshot:



Question 3

Problem Statement: Use the 3 by 3 matrix with the rows 1 2 3, 4 5 6 and 7 8 9. Print the sum of each row, one line per row, in the form Row 1 sum: followed by the total.

Source Code (Main.java):

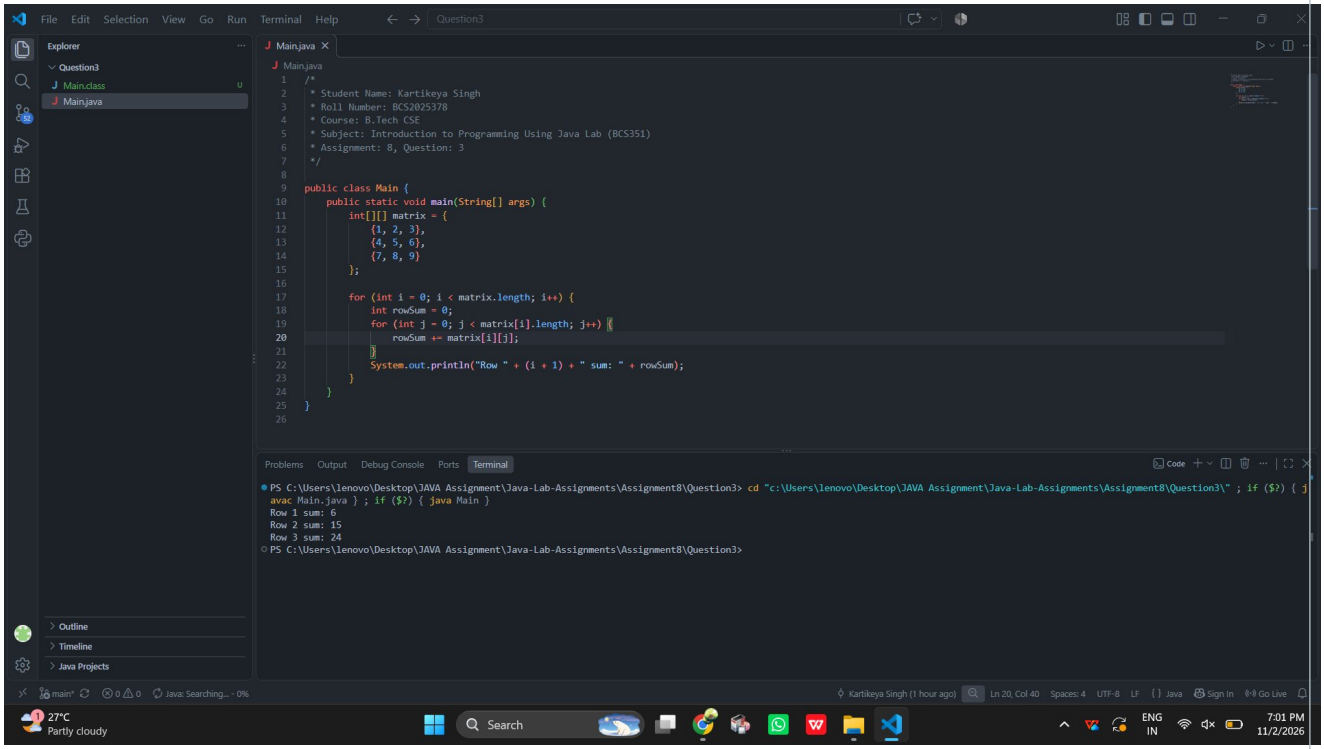
```
/*
 * Student Name: Kartikeya Singh
 * Roll Number: BCS2025378
 * Course: B.Tech CSE
```

```
 * Subject: Introduction to Programming Using Java Lab (BCS351)
 * Assignment: 8, Question: 3
 */

public class Main {
    public static void main(String[] args) {
        int[][] matrix = {
            {1, 2, 3},
            {4, 5, 6},
            {7, 8, 9}
        };

        for (int i = 0; i < matrix.length; i++) {
            int rowSum = 0;
            for (int j = 0; j < matrix[i].length; j++) {
                rowSum += matrix[i][j];
            }
            System.out.println("Row " + (i + 1) + " sum: " + rowSum);
        }
    }
}
```

Execution Output Screenshot:



Assignment 9: Matrices: Rows and Columns

Assignment Aim / Objective: Visit the rows and columns of a matrix with nested loops to print, add and count its values.

Question 1

Problem Statement: Use the 3 by 3 matrix with rows 1 2 3, 4 5 6 and 7 8 9. Print First row: and its values, then Last row: and its values, with a space between values.

Source Code (Main.java):

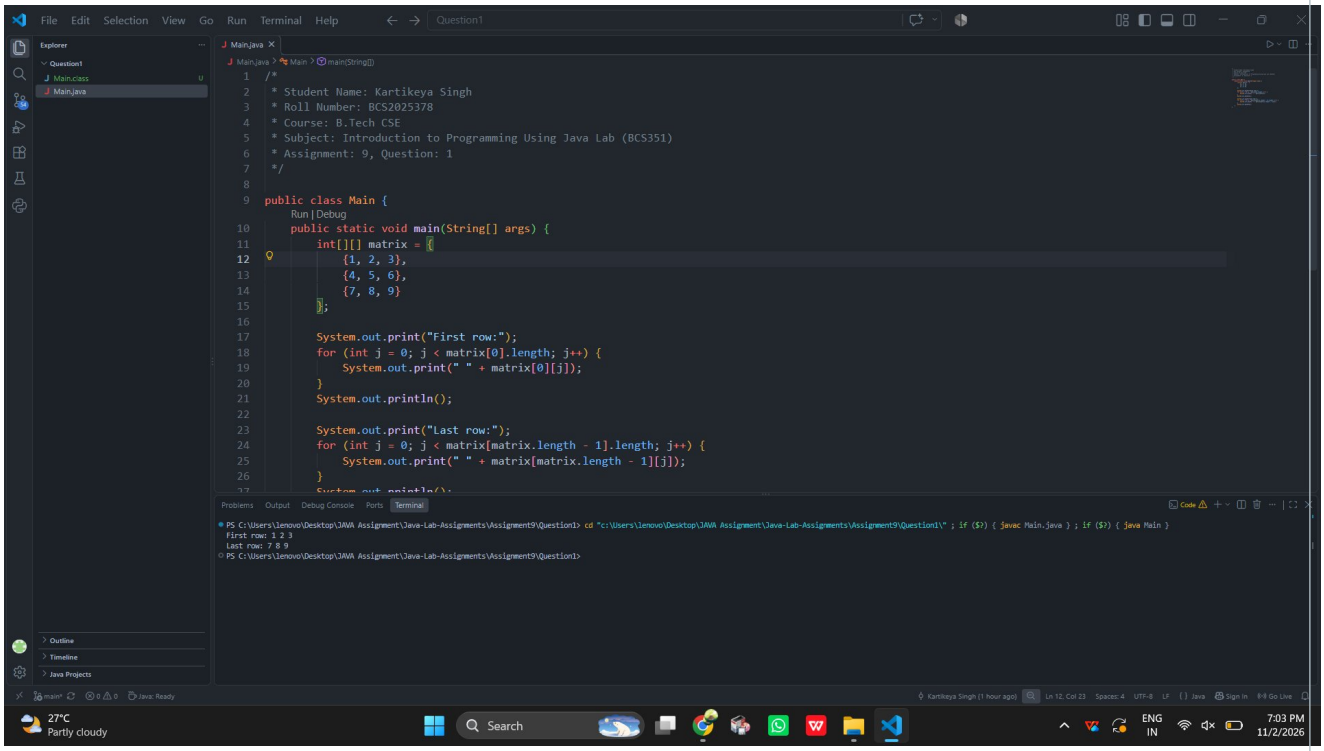
```
/*
 * Student Name: Kartikeya Singh
 * Roll Number: BCS2025378
 * Course: B.Tech CSE
 * Subject: Introduction to Programming Using Java Lab (BCS351)
 * Assignment: 9, Question: 1
 */

public class Main {
    public static void main(String[] args) {
        int[][] matrix = {
            {1, 2, 3},
            {4, 5, 6},
            {7, 8, 9}
        };

        System.out.print("First row:");
        for (int j = 0; j < matrix[0].length; j++) {
            System.out.print(" " + matrix[0][j]);
        }
        System.out.println();

        System.out.print("Last row:");
        for (int j = 0; j < matrix[matrix.length - 1].length; j++) {
            System.out.print(" " + matrix[matrix.length - 1][j]);
        }
        System.out.println();
    }
}
```

Execution Output Screenshot:



Question 2

Problem Statement: Use the 3 by 3 matrix with the rows 1 2 3, 4 5 6 and 7 8 9. Print the sum of each column, one line per column, in the form Column 1 sum: followed by the total.

Source Code (Main.java):

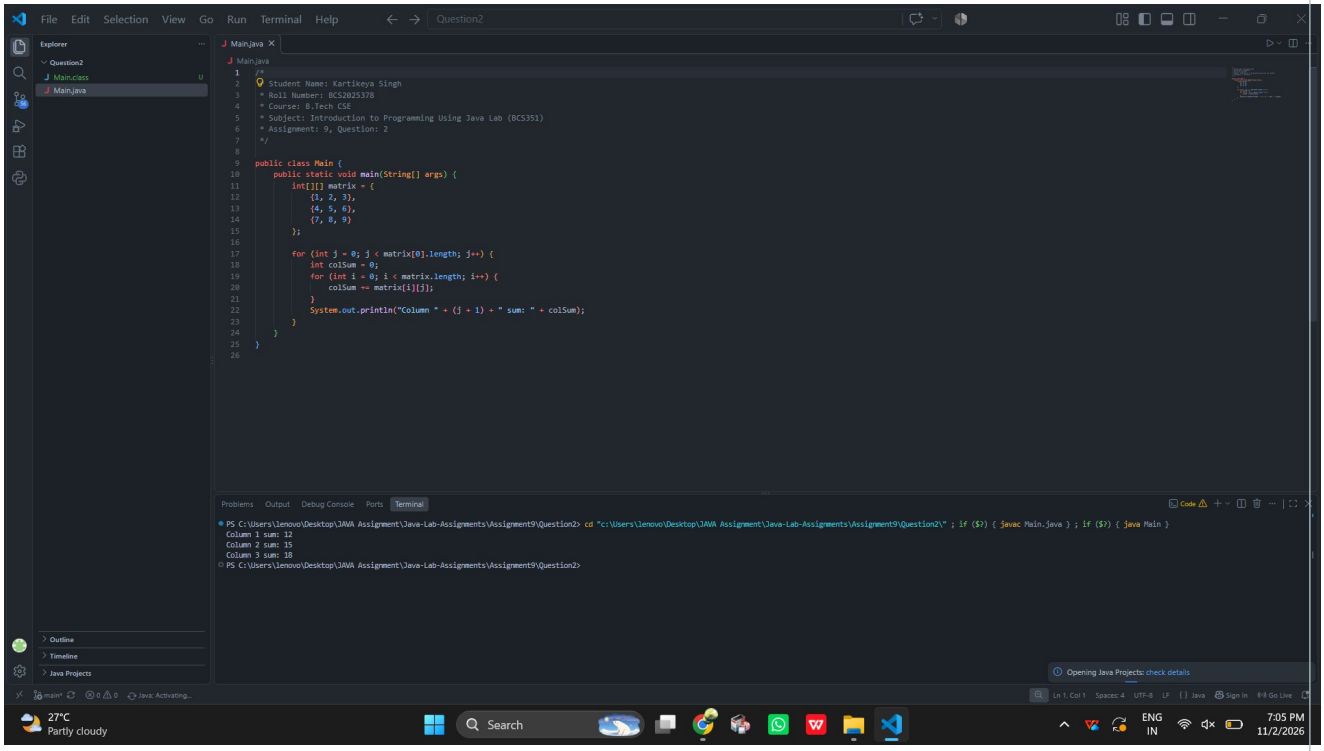
```
/*
 * Student Name: Kartikeya Singh
 * Roll Number: BCS2025378
 * Course: B.Tech CSE
 * Subject: Introduction to Programming Using Java Lab (BCS351)
 * Assignment: 9, Question: 2
 */

public class Main {
    public static void main(String[] args) {
        int[][] matrix = {
            {1, 2, 3},
            {4, 5, 6},
            {7, 8, 9}
        };

        for (int j = 0; j < matrix[0].length; j++) {
            int colSum = 0;
            for (int i = 0; i < matrix.length; i++) {
```

```
                colSum += matrix[i][j];
            }
            System.out.println("Column " + (j + 1) + " sum: " + colSum);
        }
    }
}
```

Execution Output Screenshot:



Question 3

Problem Statement: Use the 3 by 3 matrix with the rows 1 2 3, 4 5 6 and 7 8 9. Count the values greater than 5 and print Values greater than 5: followed by the count.

Source Code (Main.java):

```
/*
 * Student Name: Kartikeya Singh
 * Roll Number: BCS2025378
 * Course: B.Tech CSE
 * Subject: Introduction to Programming Using Java Lab (BCS351)
 * Assignment: 9, Question: 3
 */

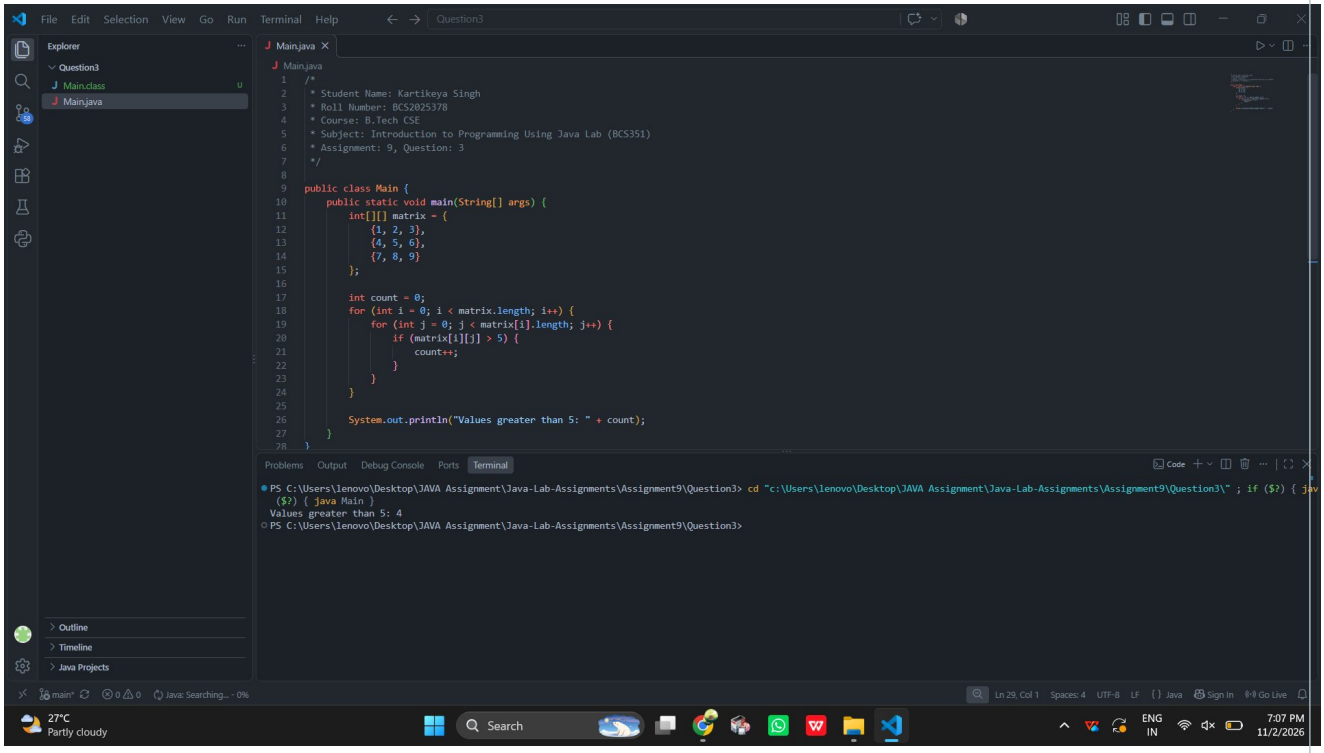
public class Main {
    public static void main(String[] args) {
        int[][] matrix = {
```

```
            {1, 2, 3},
            {4, 5, 6},
            {7, 8, 9}
        };

        int count = 0;
        for (int i = 0; i < matrix.length; i++) {
            for (int j = 0; j < matrix[i].length; j++) {
                if (matrix[i][j] > 5) {
                    count++;
                }
            }
        }

        System.out.println("Values greater than 5: " + count);
    }
}
```

Execution Output Screenshot:



Assignment 10: Switch and ArrayList

Assignment Aim / Objective: Choose a branch with a switch statement, and store and change a list of numbers in an ArrayList.

Question 1

Problem Statement: Store 3 in an int named day. Use a switch statement to print the name of the day: 1 is Monday, 2 is Tuesday and 3 is Wednesday. For any other number, print Invalid day.

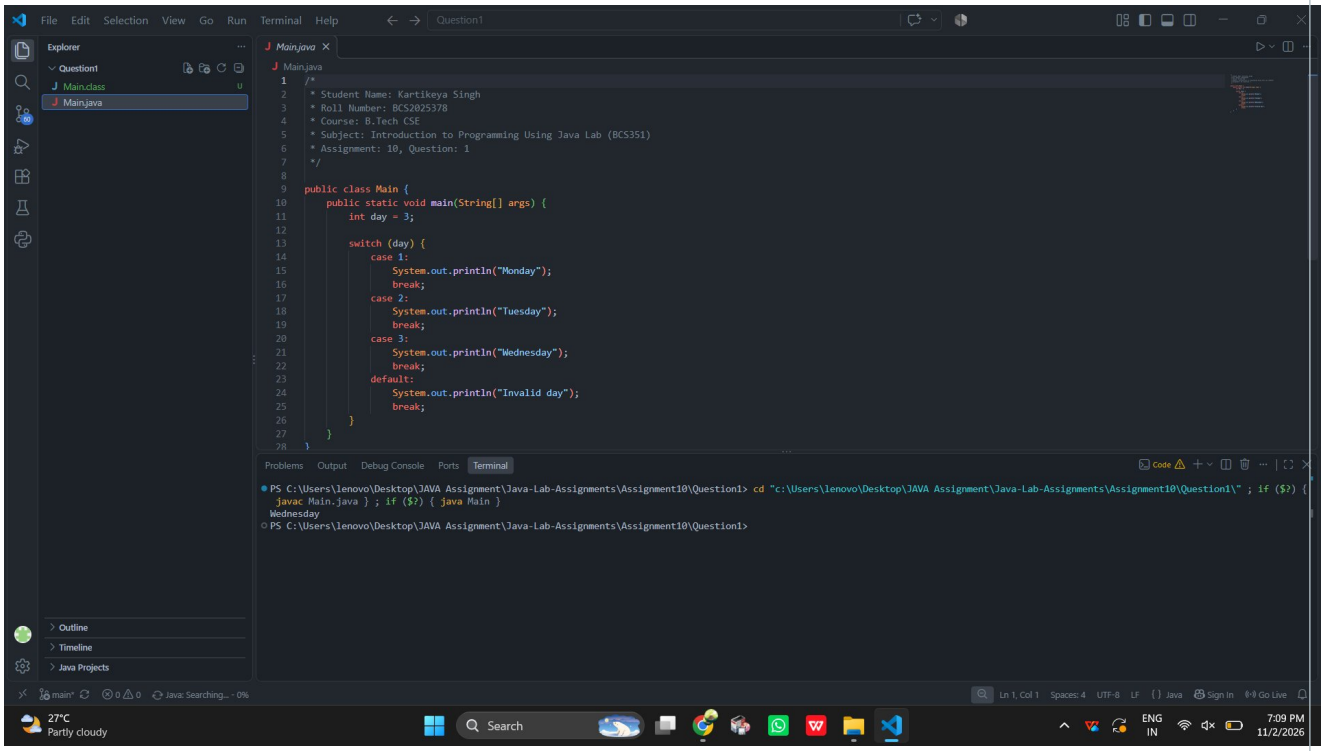
Source Code (Main.java):

```
/*
 * Student Name: Kartikeya Singh
 * Roll Number: BCS2025378
 * Course: B.Tech CSE
 * Subject: Introduction to Programming Using Java Lab (BCS351)
 * Assignment: 10, Question: 1
 */

public class Main {
    public static void main(String[] args) {
        int day = 3;

        switch (day) {
            case 1:
                System.out.println("Monday");
                break;
            case 2:
                System.out.println("Tuesday");
                break;
            case 3:
                System.out.println("Wednesday");
                break;
            default:
                System.out.println("Invalid day");
                break;
        }
    }
}
```

Execution Output Screenshot:



Question 2

Problem Statement: Create an ArrayList of Integer named list. Add the values 10, 20, 30 and 40. Print the whole list. Then print Size: followed by its size, and Item at index 2: followed by the value at index 2.

Source Code (Main.java):

```
/*
 * Student Name: Kartikeya Singh
 * Roll Number: BCS2025378
 * Course: B.Tech CSE
 * Subject: Introduction to Programming Using Java Lab (BCS351)
 * Assignment: 10, Question: 2
 */

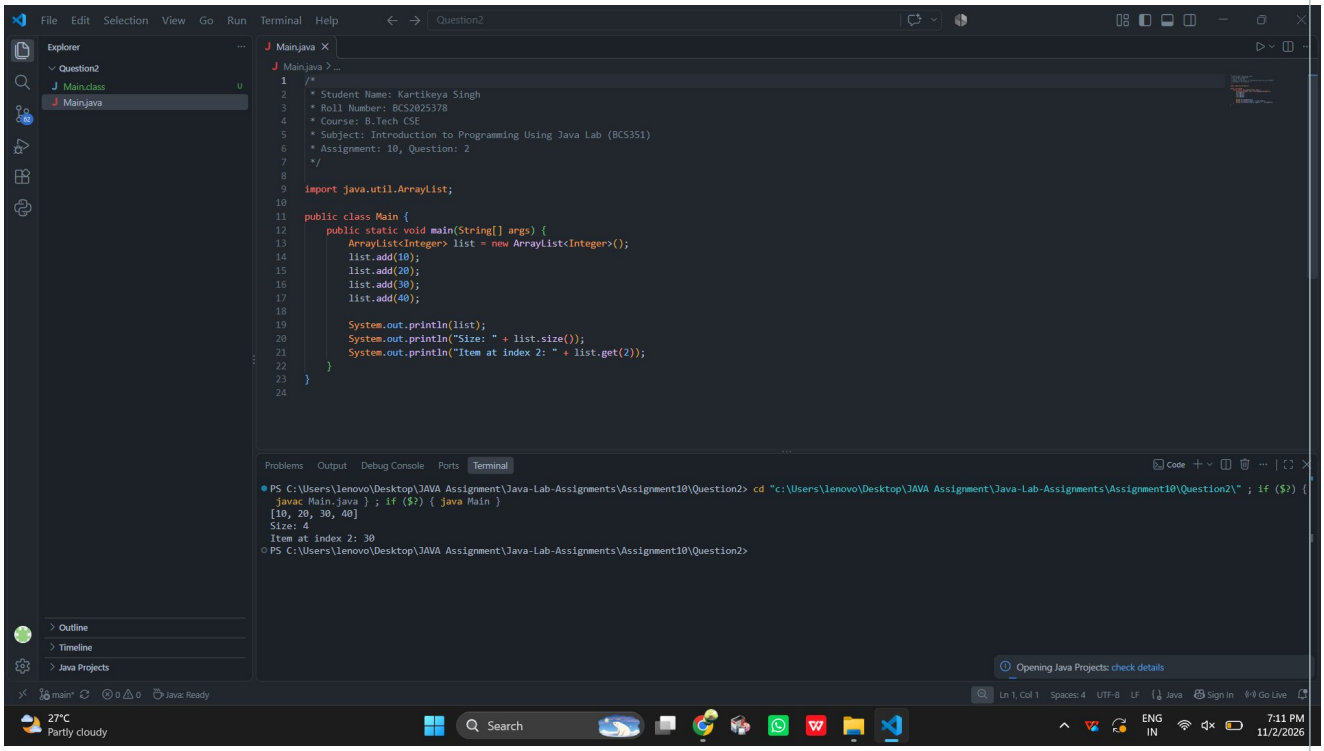
import java.util.ArrayList;

public class Main {
    public static void main(String[] args) {
        ArrayList<Integer> list = new ArrayList<Integer>();
        list.add(10);
        list.add(20);
        list.add(30);
        list.add(40);

        System.out.println(list);
```

```
        System.out.println("Size: " + list.size());
        System.out.println("Item at index 2: " + list.get(2));
    }
}
```

Execution Output Screenshot:



Question 3

Problem Statement: Create an ArrayList of Integer named list with the values 10, 20, 30 and 40. Use set to change index 1 to 25, and remove to delete index 3. Print the list and Size: followed by its size.

Source Code (Main.java):

```
/*
 * Student Name: Kartikeya Singh
 * Roll Number: BCS2025378
 * Course: B.Tech CSE
 * Subject: Introduction to Programming Using Java Lab (BCS351)
 * Assignment: 10, Question: 3
 */

import java.util.ArrayList;

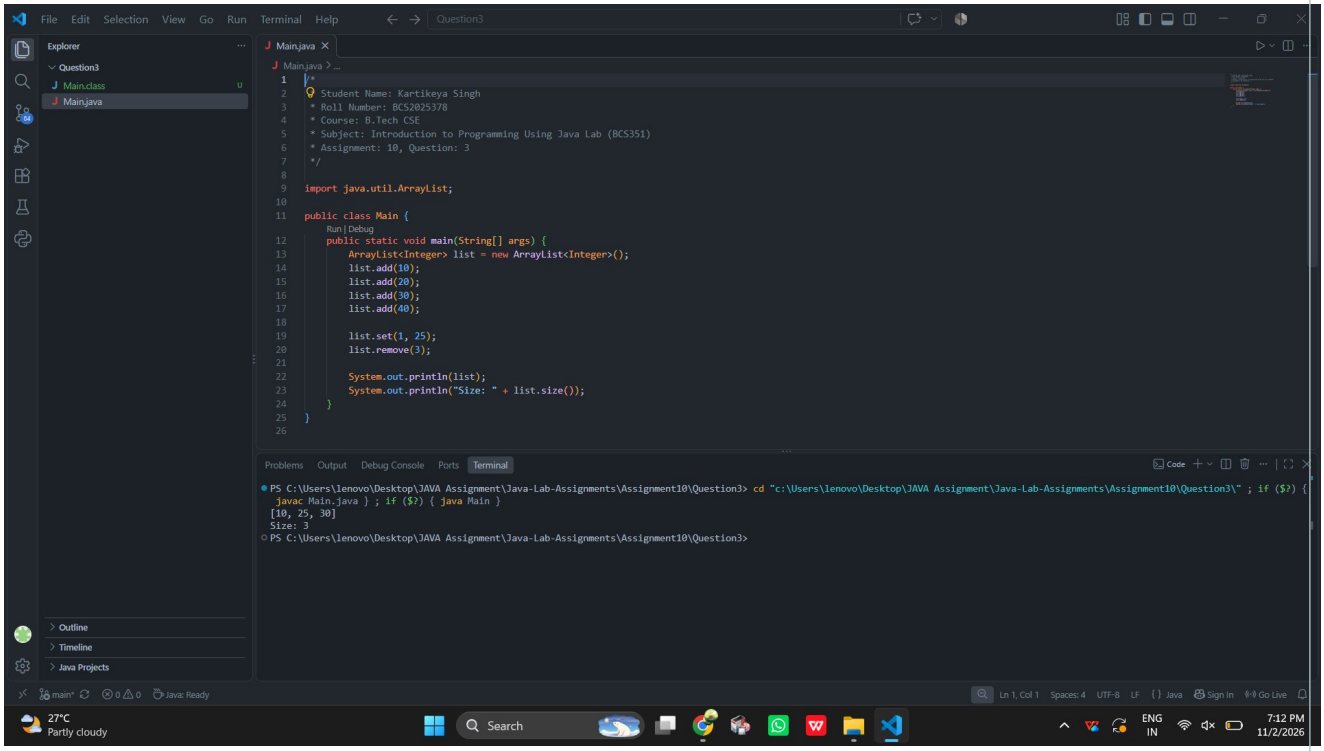
public class Main {
    public static void main(String[] args) {
        ArrayList<Integer> list = new ArrayList<Integer>();
```

```
        list.add(10);
        list.add(20);
        list.add(30);
        list.add(40);

        list.set(1, 25);
        list.remove(3);

        System.out.println(list);
        System.out.println("Size: " + list.size());
    }
}
```

Execution Output Screenshot:



Assignment 11: Bubble Sort and Selection Sort

Assignment Aim / Objective: Sort small arrays with bubble sort and selection sort, and count the swaps.

Question 1

Problem Statement: Create the int array 5, 1, 4, 2, 8. Sort it in increasing order with bubble sort. Print Sorted: followed by the sorted values in square brackets, separated by commas.

Source Code (Main.java):

```
/*
 * Student Name: Kartikeya Singh
 * Roll Number: BCS2025378
 * Course: B.Tech CSE
 * Subject: Introduction to Programming Using Java Lab (BCS351)
 * Assignment: 11, Question: 1
 */

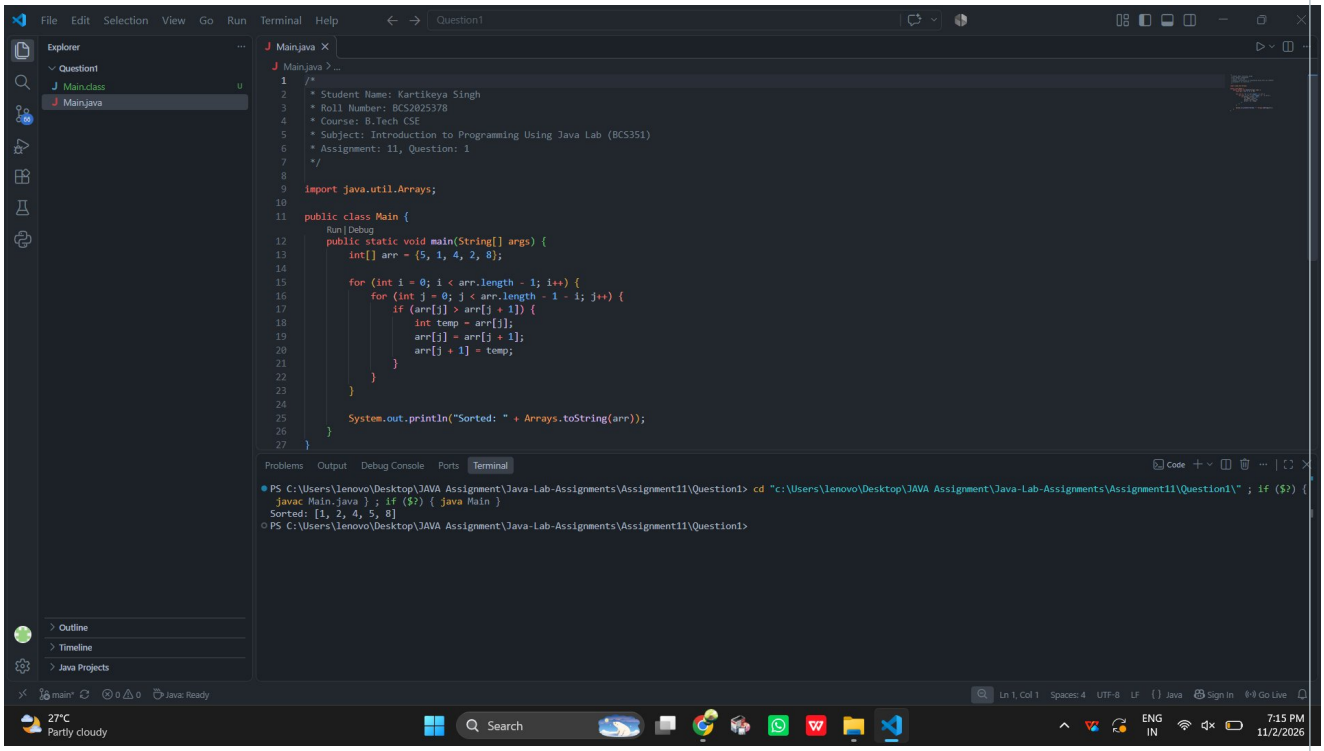
import java.util.Arrays;

public class Main {
    public static void main(String[] args) {
        int[] arr = {5, 1, 4, 2, 8};

        for (int i = 0; i < arr.length - 1; i++) {
            for (int j = 0; j < arr.length - 1 - i; j++) {
                if (arr[j] > arr[j + 1]) {
                    int temp = arr[j];
                    arr[j] = arr[j + 1];
                    arr[j + 1] = temp;
                }
            }
        }

        System.out.println("Sorted: " + Arrays.toString(arr));
    }
}
```

Execution Output Screenshot:



Question 2

Problem Statement: Create the int array 64, 25, 12, 22, 11. Sort it in increasing order with selection sort. Print Sorted: followed by the sorted values in square brackets, separated by commas.

Source Code (Main.java):

```
/*
 * Student Name: Kartikeya Singh
 * Roll Number: BCS2025378
 * Course: B.Tech CSE
 * Subject: Introduction to Programming Using Java Lab (BCS351)
 * Assignment: 11, Question: 2
 */

import java.util.Arrays;

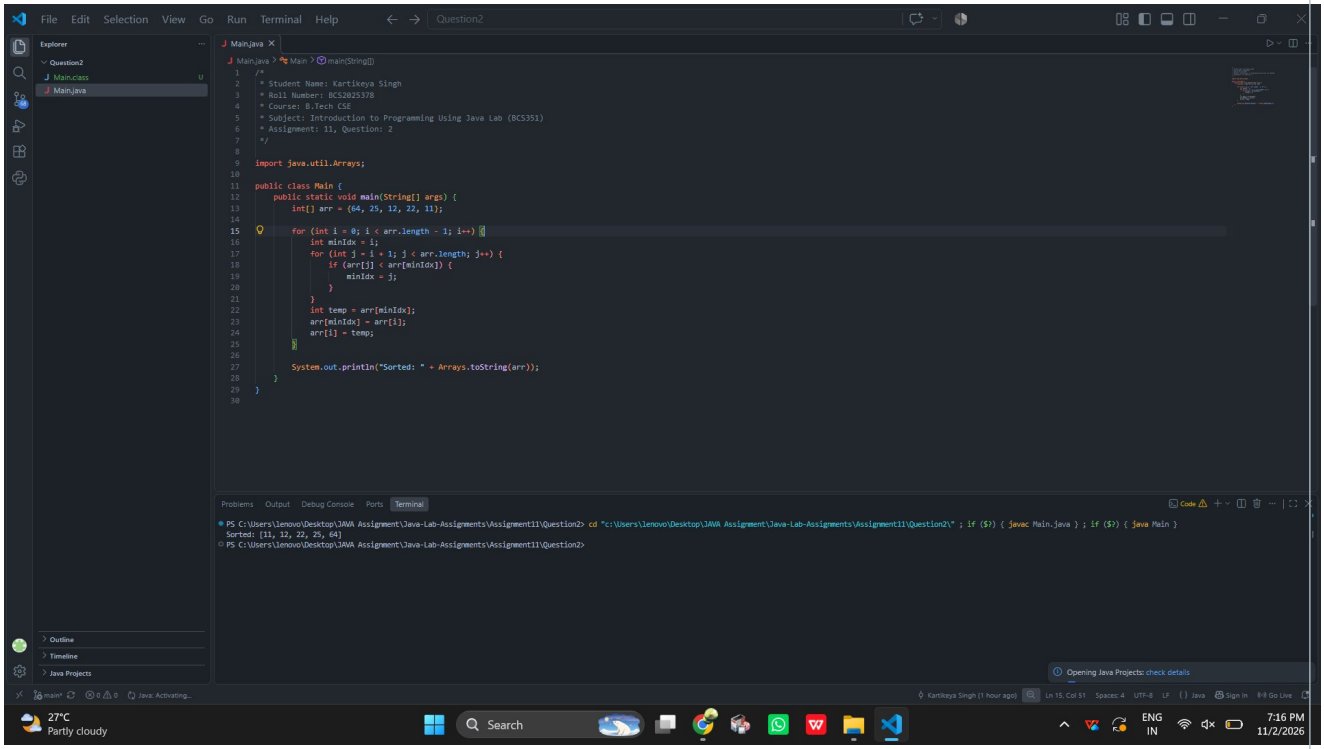
public class Main {
    public static void main(String[] args) {
        int[] arr = {64, 25, 12, 22, 11};

        for (int i = 0; i < arr.length - 1; i++) {
            int minIdx = i;
            for (int j = i + 1; j < arr.length; j++) {
                if (arr[j] < arr[minIdx]) {
                    minIdx = j;
```

```
                }
            }
            int temp = arr[minIdx];
            arr[minIdx] = arr[i];
            arr[i] = temp;
        }

        System.out.println("Sorted: " + Arrays.toString(arr));
    }
}
```

Execution Output Screenshot:



Question 3

Problem Statement: Use the array 5, 1, 4, 2, 8 and sort it with bubble sort. Count the swaps (each time two values change places). Print Swaps: followed by the count.

Source Code (Main.java):

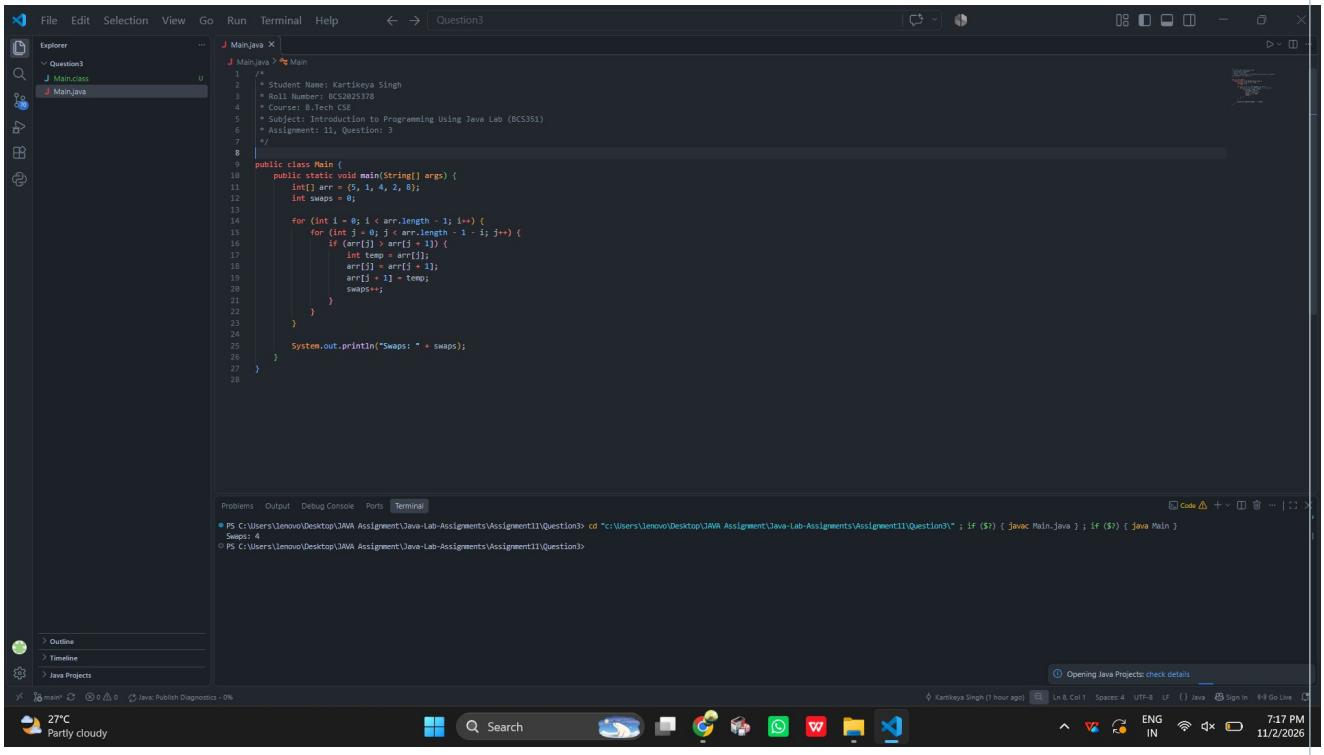
```
/*
 * Student Name: Kartikeya Singh
 * Roll Number: BCS2025378
 * Course: B.Tech CSE
 * Subject: Introduction to Programming Using Java Lab (BCS351)
 * Assignment: 11, Question: 3
 */
```

```
public class Main {
    public static void main(String[] args) {
        int[] arr = {5, 1, 4, 2, 8};
        int swaps = 0;

        for (int i = 0; i < arr.length - 1; i++) {
            for (int j = 0; j < arr.length - 1 - i; j++) {
                if (arr[j] > arr[j + 1]) {
                    int temp = arr[j];
                    arr[j] = arr[j + 1];
                    arr[j + 1] = temp;
                    swaps++;
                }
            }
        }

        System.out.println("Swaps: " + swaps);
    }
}
```

Execution Output Screenshot:



Assignment 12: Insertion Sort

Assignment Aim / Objective: Sort small arrays with insertion sort and count how many values are shifted.

Question 1

Problem Statement: Create the int array 7, 3, 5, 1. Sort it in increasing order with insertion sort. Print Sorted: followed by the sorted values in square brackets, separated by commas.

Source Code (Main.java):

```
/*
 * Student Name: Kartikeya Singh
 * Roll Number: BCS2025378
 * Course: B.Tech CSE
 * Subject: Introduction to Programming Using Java Lab (BCS351)
 * Assignment: 12, Question: 1
 */

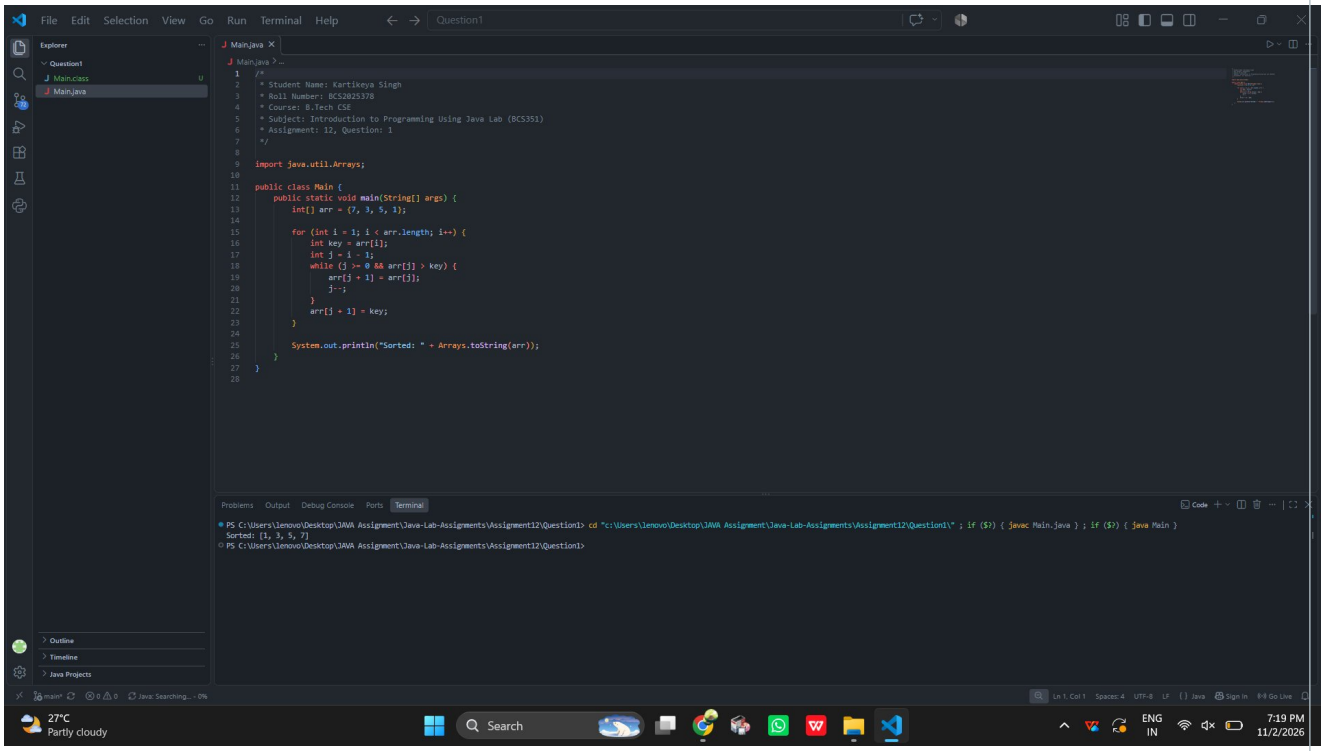
import java.util.Arrays;

public class Main {
    public static void main(String[] args) {
        int[] arr = {7, 3, 5, 1};

        for (int i = 1; i < arr.length; i++) {
            int key = arr[i];
            int j = i - 1;
            while (j >= 0 && arr[j] > key) {
                arr[j + 1] = arr[j];
                j--;
            }
            arr[j + 1] = key;
        }

        System.out.println("Sorted: " + Arrays.toString(arr));
    }
}
```

Execution Output Screenshot:



Question 2

Problem Statement: Create the int array 14, 3, 9, 21, 6. Sort it in increasing order with insertion sort. Print Sorted: followed by the sorted values in square brackets, separated by commas.

Source Code (Main.java):

```
/*
 * Student Name: Kartikeya Singh
 * Roll Number: BCS2025378
 * Course: B.Tech CSE
 * Subject: Introduction to Programming Using Java Lab (BCS351)
 * Assignment: 12, Question: 2
 */

import java.util.Arrays;

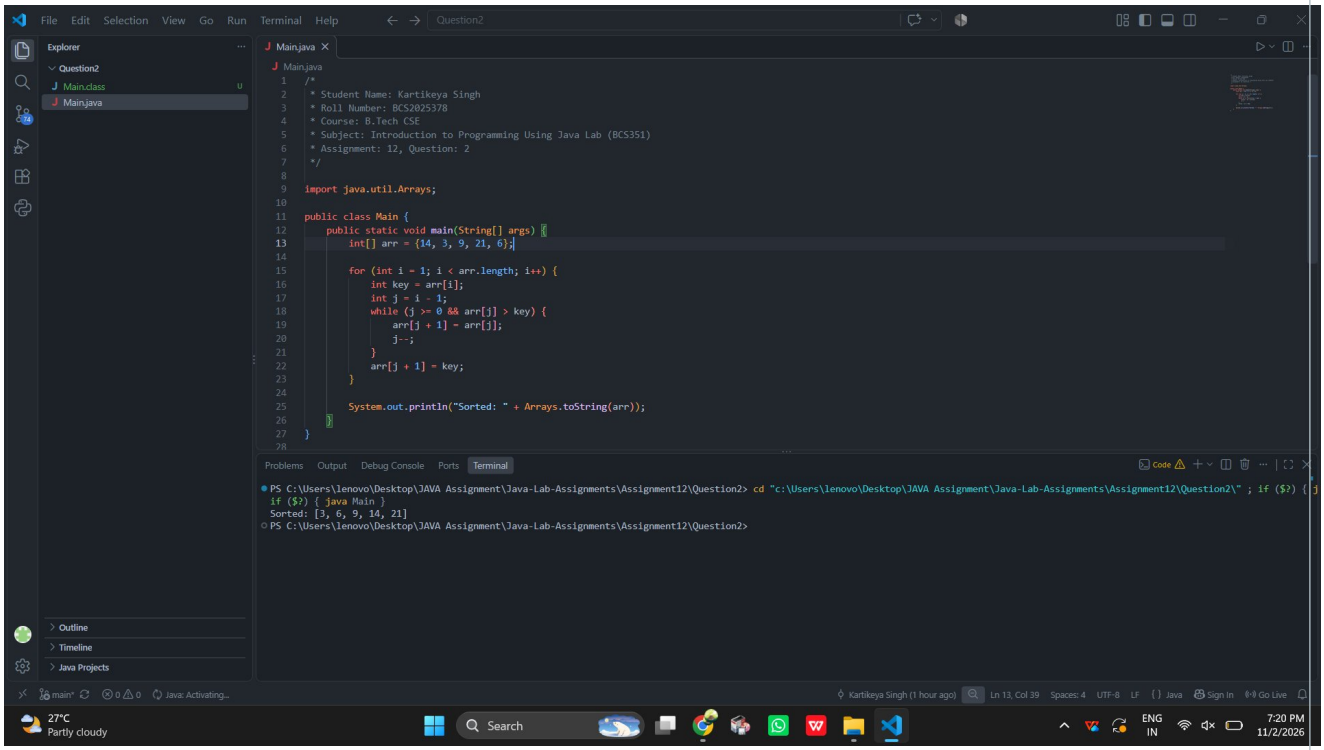
public class Main {
    public static void main(String[] args) {
        int[] arr = {14, 3, 9, 21, 6};

        for (int i = 1; i < arr.length; i++) {
            int key = arr[i];
            int j = i - 1;
            while (j >= 0 && arr[j] > key) {
                arr[j + 1] = arr[j];
```

```
                j--;
            }
            arr[j + 1] = key;
        }

        System.out.println("Sorted: " + Arrays.toString(arr));
    }
}
```

Execution Output Screenshot:



Question 3

Problem Statement: Create the int array 6, 5, 4, 3, 2, 1. Sort it in increasing order with insertion sort. Count the shifts (each time one value moves one place to the right). Print Shifts: followed by the count.

Source Code (Main.java):

```
/*
 * Student Name: Kartikeya Singh
 * Roll Number: BCS2025378
 * Course: B.Tech CSE
 * Subject: Introduction to Programming Using Java Lab (BCS351)
 * Assignment: 12, Question: 3
 */

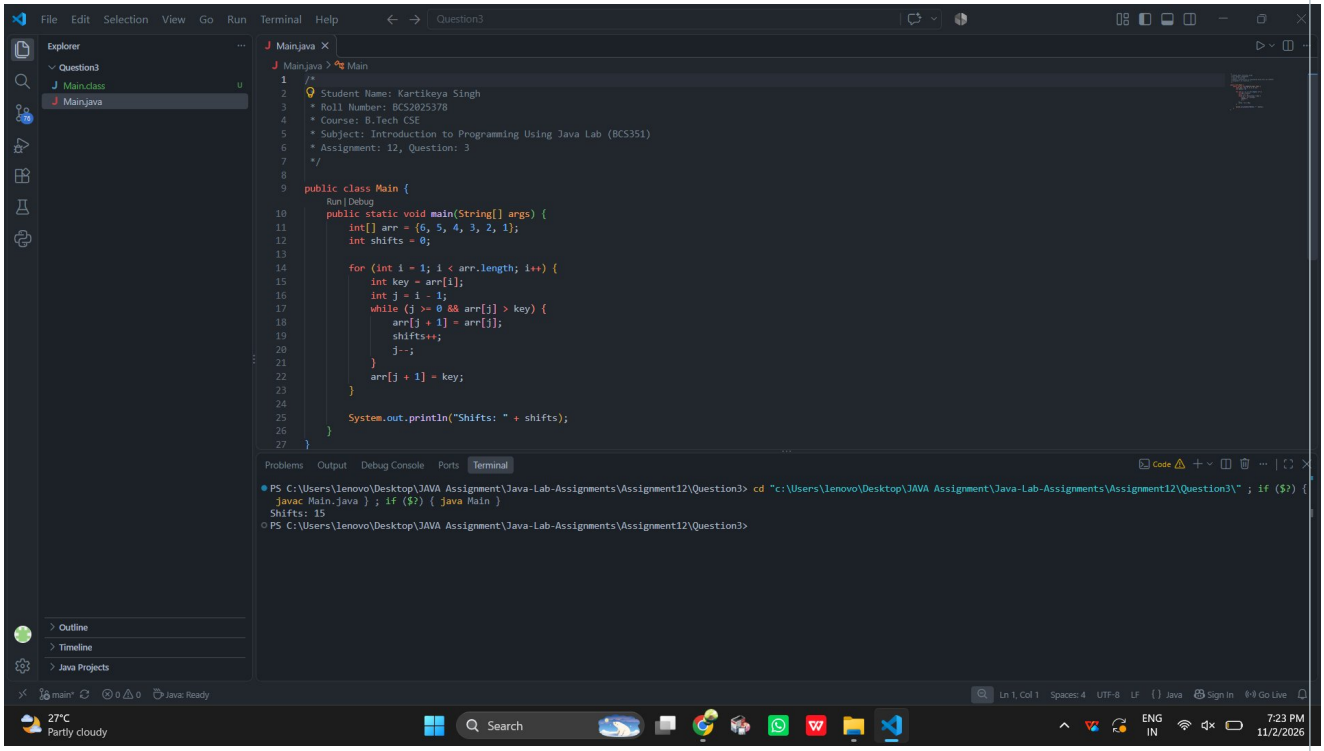
public class Main {
```

```
    public static void main(String[] args) {
        int[] arr = {6, 5, 4, 3, 2, 1};
        int shifts = 0;

        for (int i = 1; i < arr.length; i++) {
            int key = arr[i];
            int j = i - 1;
            while (j >= 0 && arr[j] > key) {
                arr[j + 1] = arr[j];
                shifts++;
                j--;
            }
            arr[j + 1] = key;
        }

        System.out.println("Shifts: " + shifts);
    }
}
```

Execution Output Screenshot:



Assignment 13: Strings: Methods and Palindromes

Assignment Aim / Objective: Use common String methods, reverse a word, check a palindrome and print substrings.

Question 1

Problem Statement: Use the word Programming. Print Length: followed by its length, Position of g: followed by the position of the first g (counting from 0), and Upper case: followed by the word in capital letters, using toUpperCase().

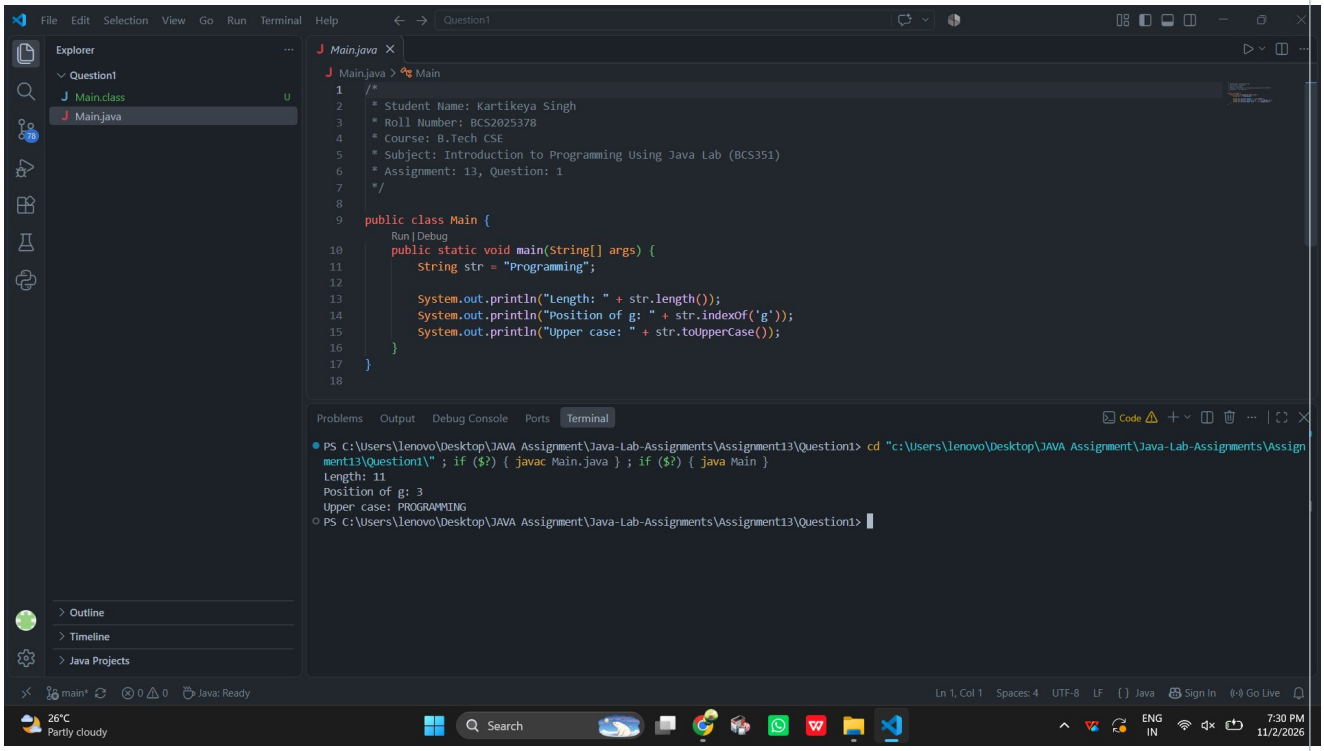
Source Code (Main.java):

```
/*
 * Student Name: Kartikeya Singh
 * Roll Number: BCS2025378
 * Course: B.Tech CSE
 * Subject: Introduction to Programming Using Java Lab (BCS351)
 * Assignment: 13, Question: 1
 */

public class Main {
    public static void main(String[] args) {
        String str = "Programming";

        System.out.println("Length: " + str.length());
        System.out.println("Position of g: " + str.indexOf('g'));
        System.out.println("Upper case: " + str.toUpperCase());
    }
}
```

Execution Output Screenshot:



Question 2

Problem Statement: Reverse the word Java. Build the reversed word one letter at a time, using charAt and + in a loop. Print Reversed: followed by the result.

Source Code (Main.java):

```
/*
 * Student Name: Kartikeya Singh
 * Roll Number: BCS2025378
 * Course: B.Tech CSE
 * Subject: Introduction to Programming Using Java Lab (BCS351)
 * Assignment: 13, Question: 2
 */

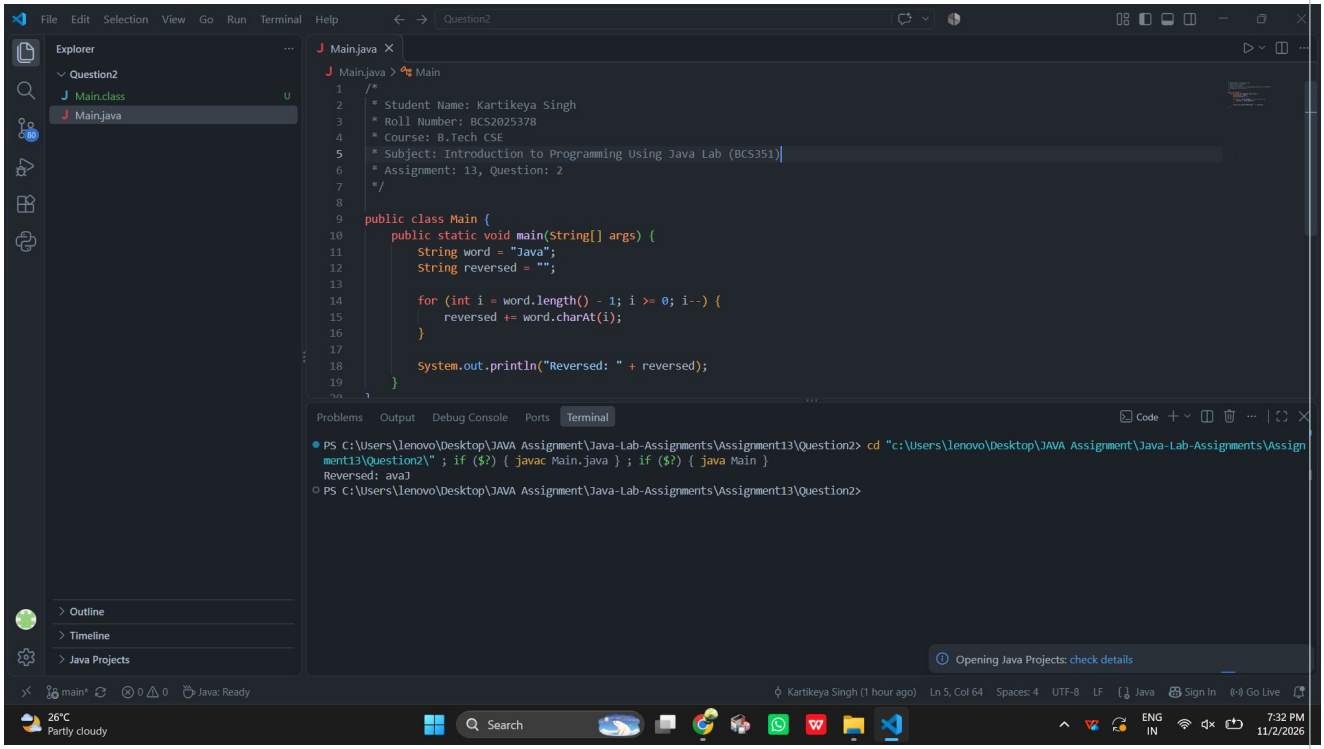
public class Main {
    public static void main(String[] args) {
        String word = "Java";
        String reversed = "";

        for (int i = word.length() - 1; i >= 0; i--) {
            reversed += word.charAt(i);
        }

        System.out.println("Reversed: " + reversed);
    }
```

```
}
```

Execution Output Screenshot:



Question 3

Problem Statement: Use the word racecar. Check whether it is a palindrome, which means it reads the same from both ends. Print the word followed by is a palindrome, or followed by is not a palindrome.

Source Code (Main.java):

```
/*
 * Student Name: Kartikeya Singh
 * Roll Number: BCS2025378
 * Course: B.Tech CSE
 * Subject: Introduction to Programming Using Java Lab (BCS351)
 * Assignment: 13, Question: 3
 */

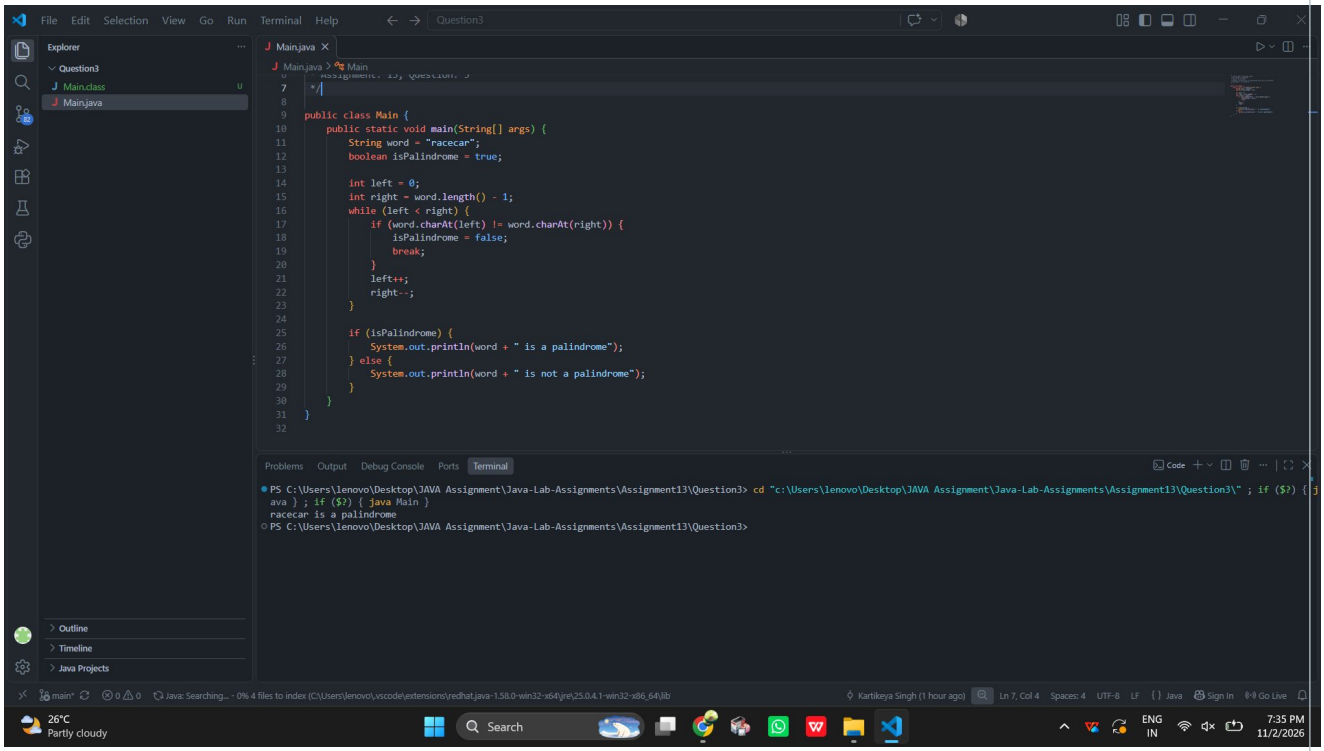
public class Main {
    public static void main(String[] args) {
        String word = "racecar";
        boolean isPalindrome = true;

        int left = 0;
        int right = word.length() - 1;
        while (left < right) {
```

```
            if (word.charAt(left) != word.charAt(right)) {
                isPalindrome = false;
                break;
            }
            left++;
            right--;
        }

        if (isPalindrome) {
            System.out.println(word + " is a palindrome");
        } else {
            System.out.println(word + " is not a palindrome");
        }
    }
}
```

Execution Output Screenshot:



Question 4

Problem Statement: Use the word abc. Print every substring of it, one on each line, using substring(). For each starting letter from left to right, print its parts from shortest to longest.

Source Code (Main.java):

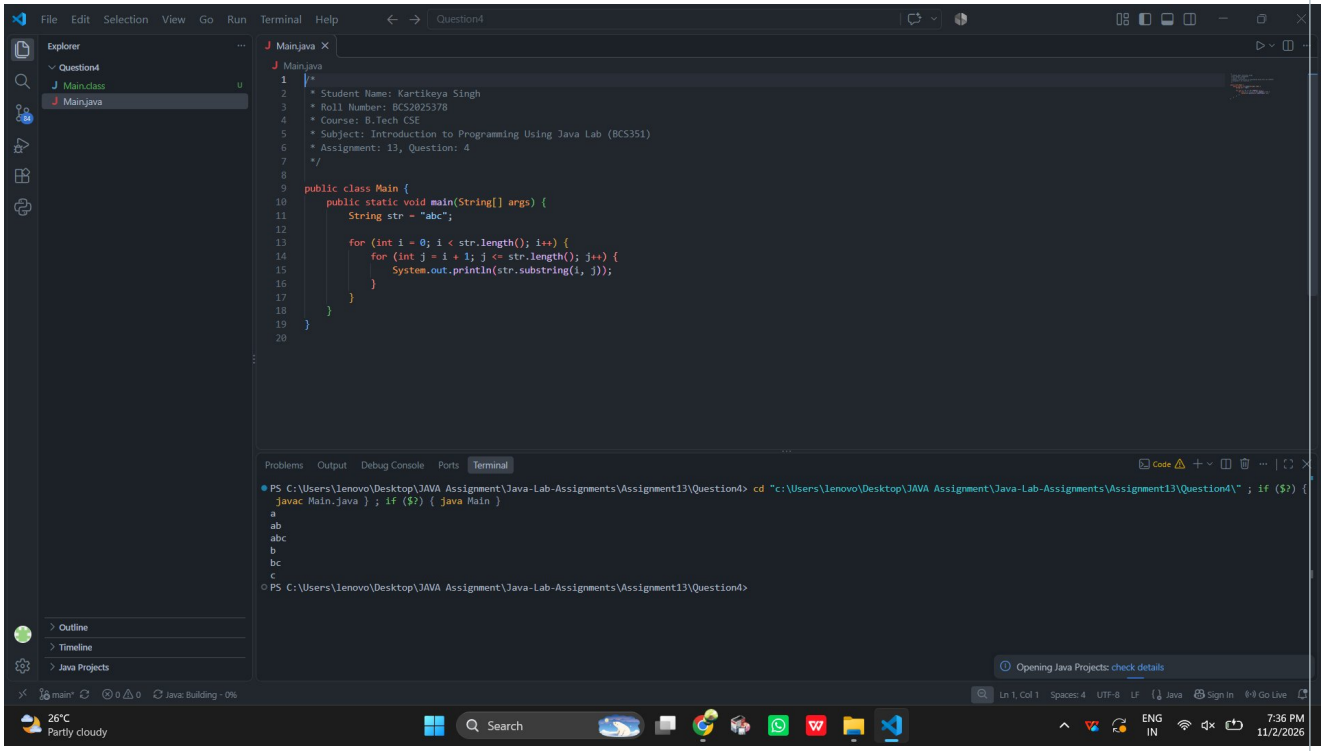
```
/*
 * Student Name: Kartikeya Singh
```

```
 * Roll Number: BCS2025378
 * Course: B.Tech CSE
 * Subject: Introduction to Programming Using Java Lab (BCS351)
 * Assignment: 13, Question: 4
 */

public class Main {
    public static void main(String[] args) {
        String str = "abc";

        for (int i = 0; i < str.length(); i++) {
            for (int j = i + 1; j <= str.length(); j++) {
                System.out.println(str.substring(i, j));
            }
        }
    }
}
```

Execution Output Screenshot:



Assignment 14: Characters: Counting and Positions

Assignment Aim / Objective: Use charAt to count vowels and a letter, and print each letter with its position.

Question 1

Problem Statement: Store programming in a String named s. Count its vowels (a, e, i, o and u). Print Vowels: followed by the count.

Source Code (Main.java):

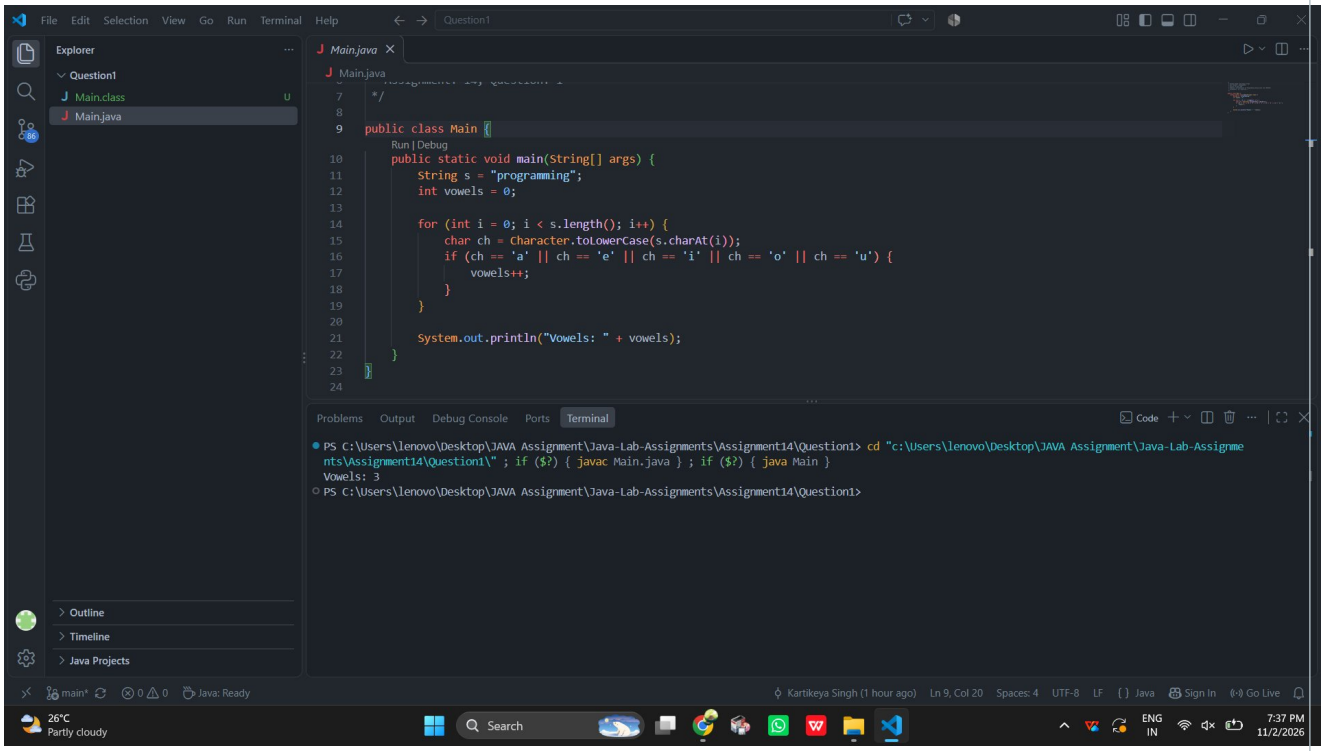
```
/*
 * Student Name: Kartikeya Singh
 * Roll Number: BCS2025378
 * Course: B.Tech CSE
 * Subject: Introduction to Programming Using Java Lab (BCS351)
 * Assignment: 14, Question: 1
 */

public class Main {
    public static void main(String[] args) {
        String s = "programming";
        int vowels = 0;

        for (int i = 0; i < s.length(); i++) {
            char ch = Character.toLowerCase(s.charAt(i));
            if (ch == 'a' || ch == 'e' || ch == 'i' || ch == 'o' || ch == 'u') {
                vowels++;
            }
        }

        System.out.println("Vowels: " + vowels);
    }
}
```

Execution Output Screenshot:



Question 2

Problem Statement: Store programming in a String named s. Count how many times the letter g appears. Print Count of g: followed by the count.

Source Code (Main.java):

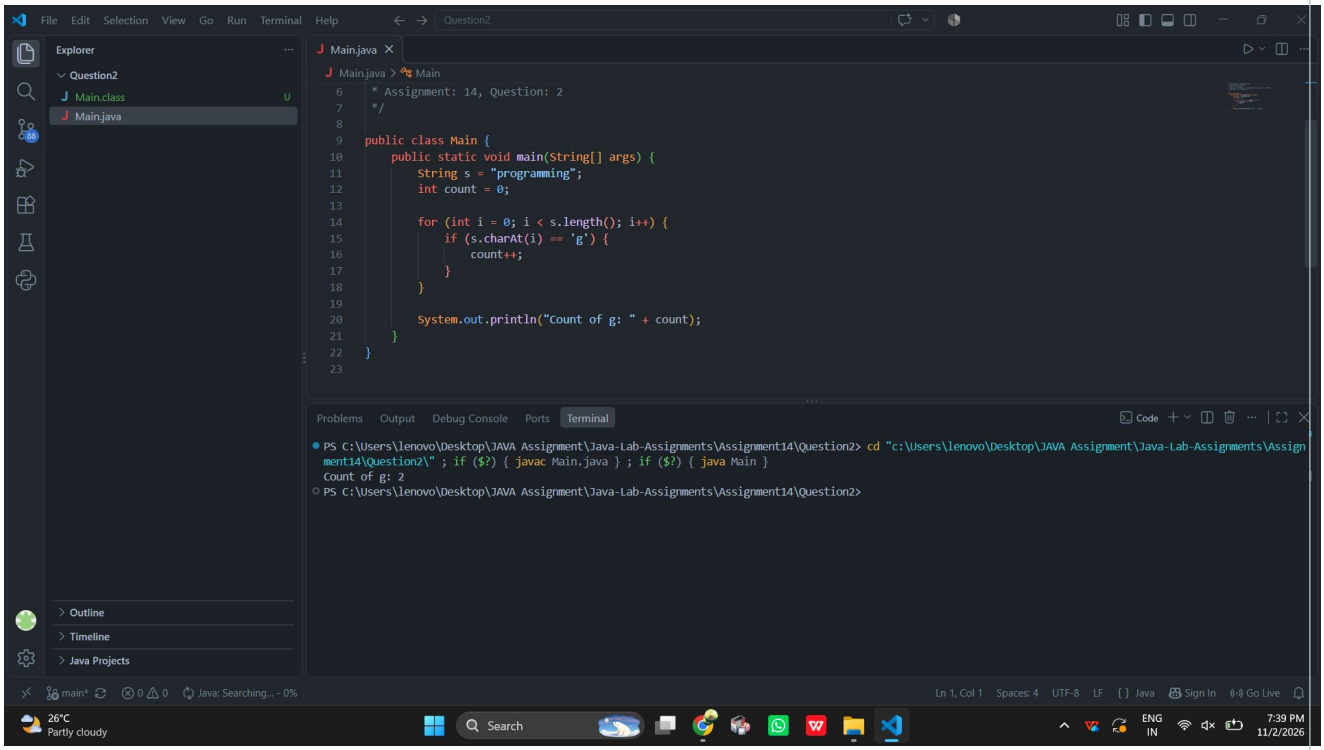
```
/*
 * Student Name: Kartikeya Singh
 * Roll Number: BCS2025378
 * Course: B.Tech CSE
 * Subject: Introduction to Programming Using Java Lab (BCS351)
 * Assignment: 14, Question: 2
 */

public class Main {
    public static void main(String[] args) {
        String s = "programming";
        int count = 0;

        for (int i = 0; i < s.length(); i++) {
            if (s.charAt(i) == 'g') {
                count++;
            }
        }
```

```
        System.out.println("Count of g: " + count);
    }
}
```

Execution Output Screenshot:



Question 3

Problem Statement: Store hello in a String named s. Print each letter on its own line as its position, a colon, a space and the letter. Positions start at 0.

Source Code (Main.java):

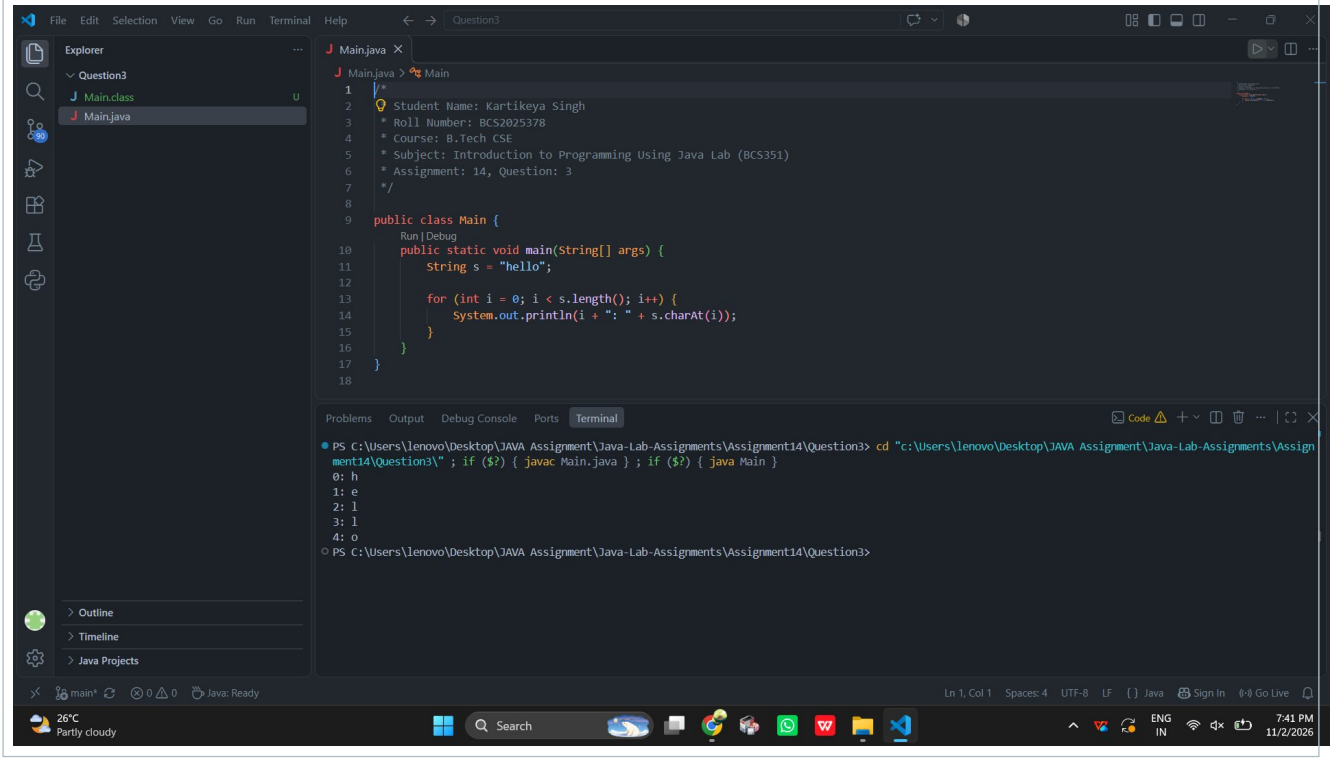
```
/*
 * Student Name: Kartikeya Singh
 * Roll Number: BCS2025378
 * Course: B.Tech CSE
 * Subject: Introduction to Programming Using Java Lab (BCS351)
 * Assignment: 14, Question: 3
 */

public class Main {
    public static void main(String[] args) {
        String s = "hello";

        for (int i = 0; i < s.length(); i++) {
            System.out.println(i + ": " + s.charAt(i));
```

```
        }
    }
}
```

Execution Output Screenshot:



Assignment 15: StringBuilder Methods

Assignment Aim / Objective: Build and change text with StringBuilder, using append, insert and deleteCharAt.

Question 1

Problem Statement: Create a StringBuilder that starts with the text Hello. Append the text " World" to it. Print the StringBuilder, then print Length: followed by its length.

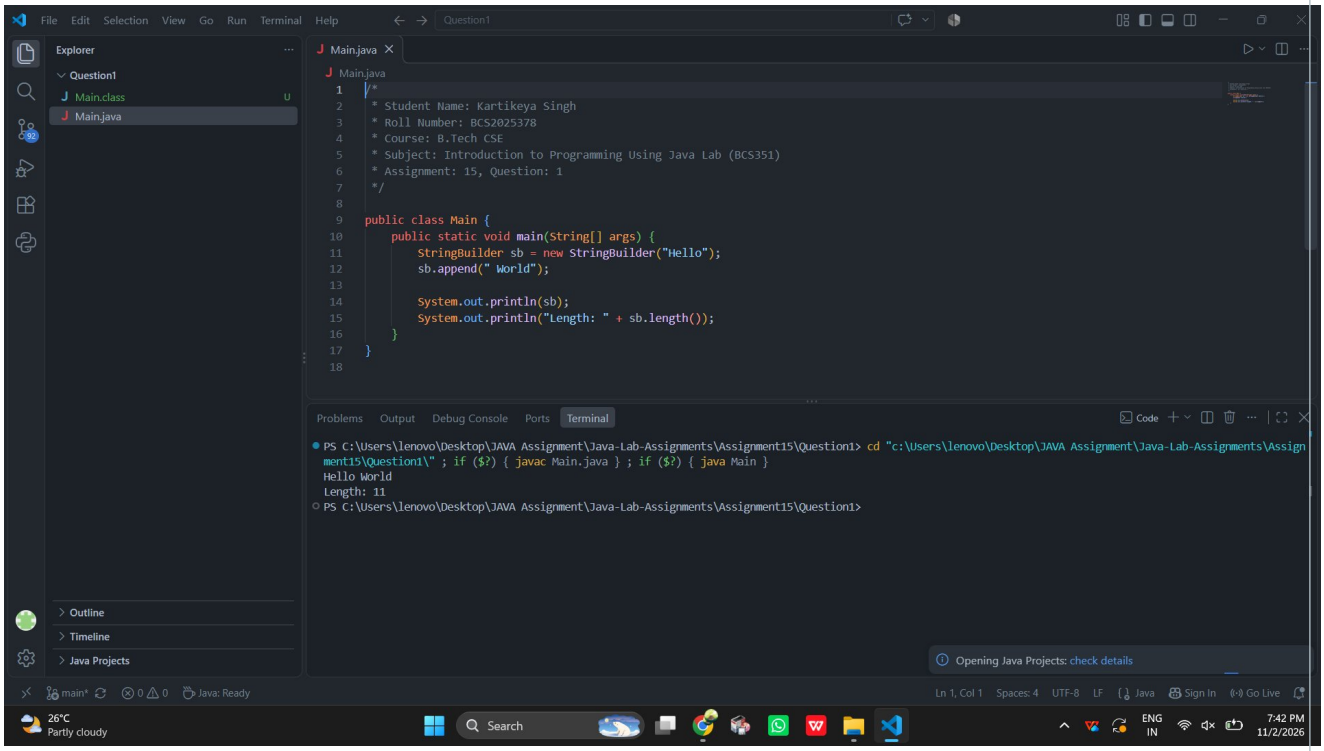
Source Code (Main.java):

```
/*
 * Student Name: Kartikeya Singh
 * Roll Number: BCS2025378
 * Course: B.Tech CSE
 * Subject: Introduction to Programming Using Java Lab (BCS351)
 * Assignment: 15, Question: 1
 */

public class Main {
    public static void main(String[] args) {
        StringBuilder sb = new StringBuilder("Hello");
        sb.append(" World");

        System.out.println(sb);
        System.out.println("Length: " + sb.length());
    }
}
```

Execution Output Screenshot:



Question 2

Problem Statement: Use a StringBuilder. Append each number from 1 to 5, followed by a comma. Then remove the last comma with deleteCharAt. Print the result.

Source Code (Main.java):

```
/*
 * Student Name: Kartikeya Singh
 * Roll Number: BCS2025378
 * Course: B.Tech CSE
 * Subject: Introduction to Programming Using Java Lab (BCS351)
 * Assignment: 15, Question: 2
 */

public class Main {
    public static void main(String[] args) {
        StringBuilder sb = new StringBuilder();

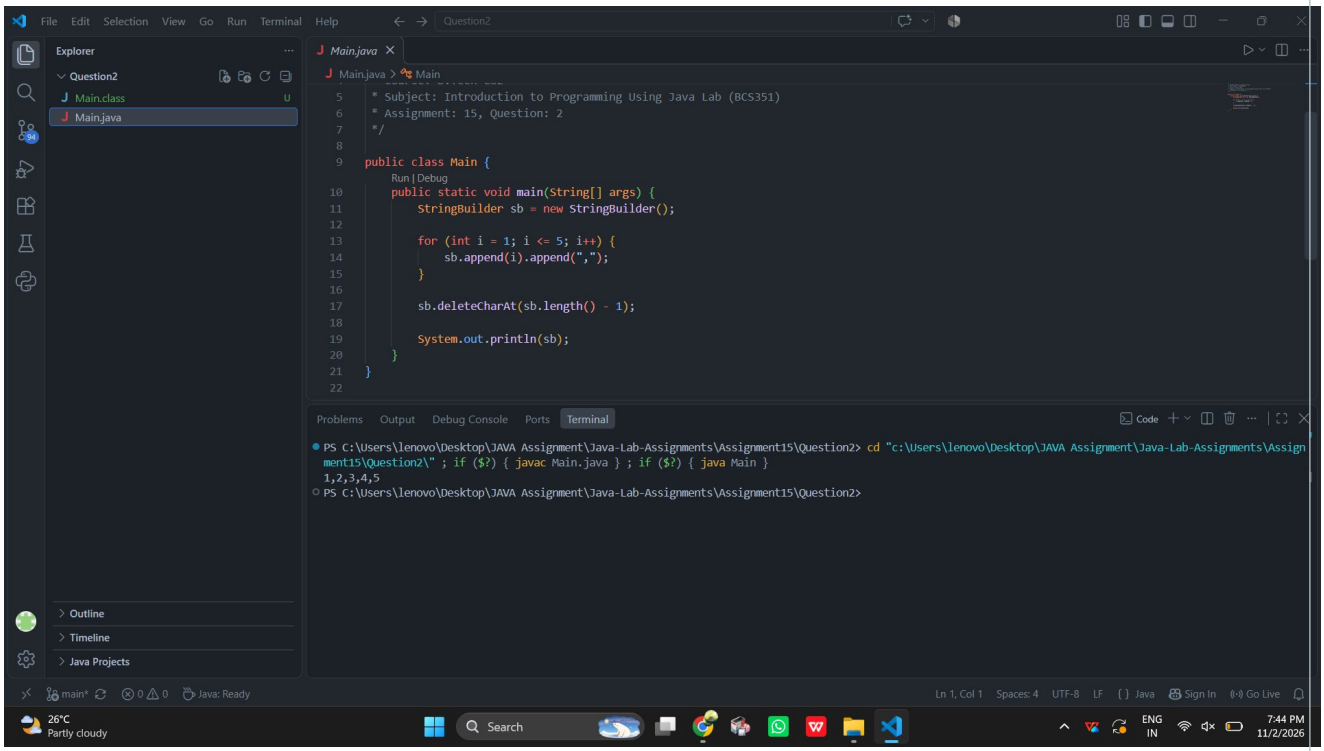
        for (int i = 1; i <= 5; i++) {
            sb.append(i).append(",");
        }

        sb.deleteCharAt(sb.length() - 1);

        System.out.println(sb);
```

```
    }
}
```

Execution Output Screenshot:



Question 3

Problem Statement: Create a StringBuilder that starts with the text Java. Use insert to add the text Hello followed by a space at index 0. Print the StringBuilder.

Source Code (Main.java):

```
/*
 * Student Name: Kartikeya Singh
 * Roll Number: BCS2025378
 * Course: B.Tech CSE
 * Subject: Introduction to Programming Using Java Lab (BCS351)
 * Assignment: 15, Question: 3
 */

public class Main {
    public static void main(String[] args) {
        StringBuilder sb = new StringBuilder("Java");
        sb.insert(0, "Hello ");

        System.out.println(sb);
    }
```

```
}
```

Execution Output Screenshot:

